

Commodity Prices, Shadow Wages, and Household Labor Allocation under Imperfect Labor Markets

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July 31, 2026

Abstract

Variation in local poultry prices can influence household labor allocation, yet little is known about the mechanisms through which these effects arise when rural labor markets are imperfect. Using three rounds of the Bangladesh Integrated Household Survey (2011–2019), this paper examines how local poultry prices affect labor allocation among poultry-producing households. Guided by the non-separable agricultural household model, I recover household-specific shadow wages and household production resources from an estimated household production function and examine how these mechanisms shape labor allocation using an instrumental-variable approach. The results show that local poultry prices primarily affect women’s labor allocation. Higher household-specific shadow wages increase labor devoted to poultry production, whereas greater household production resources are associated with broader adjustments in overall labor supply. These findings show that variation in local poultry prices influences household labor allocation through household-specific opportunity costs rather than observed market wages, highlighting the importance of accounting for labor-market imperfections when evaluating livestock market development and agricultural price policies.

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1 Introduction

Agricultural prices play a central role in shaping rural livelihoods in developing countries. Changes in output prices affect farm profitability, household income, and the returns to family labor, prompting farm households to adjust production and labor allocation in response to changing economic incentives. Understanding these responses is essential for evaluating agricultural commercialization, market integration, and the welfare consequences of agricultural policies (Bellemare and Wichman, 2020; Lederman and Porto, 2016). In smallholder farming systems, where family labor remains the primary production input through which households respond to price changes in agriculture markets (Janvry and Sadoulet, 1992; Dorward, 2013; Singh et al., 1986).

How households respond to changes in local agricultural prices depends on the functioning of rural factor markets. When labor markets are imperfect, higher output prices increase both the return to family labor and household production resources, generating potentially offsetting substitution and income effects. As a result, household labor allocation is influenced by household-specific opportunity costs rather than observed market wages alone. Understanding these opportunity costs is therefore essential for explaining how variation in local agricultural prices translates into household labor responses.

Smallholder poultry production provides a useful setting for studying these mechanisms because production relies primarily on family labor and output is marketed through local village markets. In rural Bangladesh, women, who face more limited opportunities in formal labor markets, perform much of the routine poultry husbandry, while men are more involved in marketing and investment decisions. Consequently, variation in local chicken and egg prices may alter household production incentives and the opportunity cost of family labor differently for women and men. These institutional features make Bangladesh's poultry sector a useful setting for examining how local output prices are transmitted through household production decisions to gender-specific labor allocation.

A substantial literature has examined how agricultural output prices influence household production, labor allocation, and rural welfare in developing countries. Changes in output prices affect production incentives and household resources, leading farm households to adjust production, technology adoption, and labor use. Most of this evidence, however, comes from reduced-form analyses that relate agricultural price variation directly to household outcomes (e.g., Benjamin (1992); Miller and Urdinola (2010); Beck et al. (2019); Danzer and Grundke (2020)). These studies provide important evidence that agricultural price variation influences household production, labor allocation, and household welfare, but they reveal relatively little about the mechanisms underlying these responses. In particular, reduced-form estimates capture the combined effects of changes in the return to family labor and household production resources, making it difficult to distinguish the substitution and income mechanisms through which output-price variation influences labor allocation.

The agricultural household model offers a natural framework for separating these mechanisms. Under incomplete labor markets, production and consumption decisions become non-separable, and household labor allocation is governed by household-specific shadow wages rather than observed market wages (Janvry

and Sadoulet, 1992; Rosenzweig, 1980; Singh et al., 1986). Building on this framework, Jacoby (1993) and Skoufias (1994) recover household-specific shadow wages from the marginal value product of family labor and demonstrate their importance for explaining labor allocation under market imperfections. Subsequent studies extend this approach to environments characterized by labor, land, and credit market constraints (e.g., Abdulai and Regmi (2000); Barrett et al. (2008); Makhdum and Kousar (2021)). Despite these advances, relatively little empirical evidence has examined whether variation in local agricultural prices is transmitted through the household-specific opportunity costs implied by the agricultural household model.

This gap is especially relevant for livestock production. Most evidence on agricultural price transmission comes from staple crop agriculture, even though livestock has become one of the fastest-growing agricultural subsectors in many developing countries. In Bangladesh and many other developing economies, poultry production differs from most crop enterprises in several important respects. It relies primarily on family labor, generates multiple marketable outputs, and exhibits a pronounced gender division of labor, with women responsible for much of the routine husbandry and management. These characteristics suggest that variation in local chicken and egg prices may alter the opportunity cost of family labor and household production resources differently for men and women. Yet relatively little empirical evidence has examined these mechanisms in smallholder livestock production.

To identify the mechanisms through which variation in local poultry prices affects household labor allocation, the empirical analysis builds on the agricultural household framework of Singh et al. (1986), Jacoby (1993), and Skoufias (1994). The empirical strategy proceeds in three stages. First, I estimate the reduced-form relationship between local chicken and egg prices and the labor supply of male and female household members to establish whether variation in local poultry prices is associated with changes in household labor allocation.

Second, I recover household-specific shadow wages and household production resources from an estimated household production function. Following Jacoby (1993) and Skoufias (1994), shadow wages are obtained from the marginal value product of family labor, while household production resources combine production surplus with exogenous household income. These household-specific measures summarize the economic incentives influencing the labor allocation in non-separable agricultural households. I then examine whether variation in local poultry prices affects labor allocation through changes in household-specific shadow wages or household production resources, thereby identifying the substitution and income mechanisms predicted by the agricultural household model.

Identification exploits spatial and temporal variation in village-level chicken and egg prices observed over three rounds of the Bangladesh Integrated Household Survey. Household fixed effects account for time-invariant differences across households, while survey-round fixed effects absorb aggregate shocks common to all households. Together, this empirical strategy traces how variation in local poultry prices is transmitted through household-specific opportunity costs to gender-specific labor allocation under imperfect rural labor markets.

Four findings emerge from the analysis. First, the estimated production function shows that family labor

is a key input in household poultry production, with women contributing more to production at the margin than men. Consistent with the agricultural household model, the separability tests reject the hypothesis that observed market wages adequately measure the opportunity cost of family labor, supporting the use of household-specific shadow wages in the subsequent analysis.

Second, variation in local poultry prices affects household labor allocation, particularly among women. The reduced-form estimates show that changes in local chicken and egg prices are associated with significant adjustments in labor force participation and hours worked, although the responses differ across products and labor activities. These patterns suggest that the relationship between agricultural prices and household labor allocation operates through multiple economic mechanisms rather than a single reduced-form effect.

Third, the instrumental-variable analysis identifies the mechanisms underlying these labor responses. Higher household-specific shadow wages increase labor devoted to poultry production, consistent with substitution effects operating through changes in the return to family labor. In contrast, greater household production resources are associated primarily with broader adjustments in overall labor supply. Together, these findings indicate that variation in local poultry prices influences household labor allocation through household-specific opportunity costs rather than observed market wages alone.

Finally, the main conclusions are robust across a range of alternative specifications, including a broader estimation sample and an alternative translog production function. Although the translog specification produces similar shadow measures, the corresponding instrumental-variable estimates are weakly identified, supporting the Cobb–Douglas specification as the preferred empirical model. Overall, the findings demonstrate that accounting for household-specific opportunity costs provides a richer understanding of how variation in local poultry prices shapes gender-specific labor allocation under imperfect rural labor markets.

This paper contributes to the literature in three ways. First, it advances the literature on agricultural prices and household labor allocation by moving beyond reduced-form estimates to identify the mechanisms through which local poultry prices influence household labor decisions under imperfect labor markets. Existing studies have largely examined either the reduced-form effects of agricultural price variation on household outcomes or the role of household-specific shadow wages in non-separable agricultural households ([Beck et al., 2019](#); [Jacoby, 1993](#); [Skoufias, 1994](#)). This paper brings these two strands together by showing how local chicken and egg prices are transmitted through household-specific opportunity costs to shape labor allocation.

Second, the paper provides direct evidence on the substitution and income mechanisms predicted by the agricultural household model. By recovering household-specific shadow wages and household production resources from an estimated production function and combining them with variation in village-level chicken and egg prices, the analysis distinguishes the effects of changes in the return to family labor from changes in household production resources.

Third, the paper contributes to the literature on gender and livestock production by examining labor adjustment within Bangladesh’s smallholder poultry sector, where family labor remains the dominant production input and women play a central role in production ([Davis et al., 2009](#); [Nath et al., 2025](#)).

The findings also have broader implications for agricultural policy. Policies that improve livestock market performance, strengthen value chains, or alter producer incentives may affect household labor allocation not only by changing household production resources but also by altering the opportunity cost of family labor. Ignoring these household-specific incentives may therefore understate how agricultural price policies influence labor allocation and the distribution of work within farm households, particularly in production systems where women contribute a large share of family labor. Taken together, the results highlight the importance of accounting for labor-market imperfections when evaluating policies aimed at promoting agricultural commercialization and rural development.

The remainder of the paper is organized as follows. Section 2 describes Bangladesh’s poultry sector, section 3 presents the conceptual framework. Section 4 introduces the data and construction of local poultry prices. Section 5 outlines the empirical strategy. Section 6 presents the empirical results, followed by robustness analyses. Section 7 concludes.

2 Institutional Background and Poultry Sector Context

2.1 Growth of the Poultry Sector in Bangladesh

Bangladesh’s poultry sector has expanded rapidly over the past three decades, driven by sustained economic growth, rising household incomes, urbanization, and changing food preferences that have increased demand for animal-source foods, particularly poultry meat and eggs (Ecker and Comstock, 2021). Per capita consumption of animal-source foods increased from approximately 15 grams per person per day in 1995–96 to more than 50 grams in 2022, with poultry products accounting for much of this growth (BBS, 2022). Compared with other livestock products, poultry remains relatively affordable and has shorter production cycles, making it an important source of animal protein for Bangladeshi households.

Growth in demand has been accompanied by a rapid expansion of domestic production. Because Bangladesh imports very little poultry meat or eggs, rising consumption has been met almost entirely by domestic producers. Investments in hatcheries, feed manufacturing, veterinary services, and marketing infrastructure have accelerated the commercialization of the sector since the early 2000s. Today, poultry accounts for approximately one-quarter of national meat production, while poultry meat and egg production have grown by roughly 8–10 percent annually over the past two decades (BBS, 2023; DLS, 2021; Hashem et al., 2024). The sector has consequently become an important source of rural employment, household income, and improved household nutrition, while creating new market opportunities for smallholder producers.

Despite rapid commercialization, poultry production remains overwhelmingly smallholder-based. Millions of rural households continue to raise poultry using family labor, with women performing much of the routine husbandry and day-to-day flock management (Davis et al., 2009; Nath et al., 2025). At the same time, poultry products are marketed primarily through local village markets, where prices vary across locations and over time. Because labor markets remain imperfect in many rural areas, households rely heavily on family labor to respond to changing production incentives. As households both produce and market chickens and eggs, variation in local poultry prices directly affects the profitability of production and the opportunity cost of family labor. Together, these institutional features make Bangladesh’s poultry sector a useful setting for

examining how variation in local poultry prices is transmitted through household production decisions to gender-specific labor allocation under imperfect rural labor markets.

2.2 Structure of Smallholder Poultry Production and Gender Division of Labor

Despite rapid commercialization and the expansion of poultry enterprises, poultry production in Bangladesh remains predominantly organized around smallholder households, particularly in rural areas, where family-managed backyard and semi-commercial flocks continue to play a central role in production and marketing (Hashem et al., 2024; Hennessey et al., 2021, 2026). Rural households typically raise poultry using family labor and relatively simple production technologies. Compared with many crop enterprises, poultry production requires relatively modest capital investment and can be readily combined with other farm and household activities. Unlike specialized commercial poultry operations, smallholder households generally do not maintain separate flocks exclusively for meat or egg production. Instead, chickens are raised as a mixed household enterprise. Hens produce eggs throughout their productive life before eventually being sold or consumed for meat, while chicks are retained, sold, or raised to replenish the flock. Consequently, chicken meat and eggs are jointly produced within the same household production system, and decisions regarding flock management, labor allocation, and marketing are closely linked across the two outputs (Dolberg, 2008; Rimi et al., 2016). Smallholder systems continue to account for a substantial share of national poultry meat and egg production (Dolberg, 2008; Hennessey et al., 2021; Rimi et al., 2016; Uddin et al., 2015). These institutional characteristics closely integrate production and household decision-making, making the sector well suited for analysis within the agricultural household framework.

Household poultry production is also characterized by a distinct gender division of labor. Women perform most routine husbandry activities, including feeding, watering, cleaning poultry shelters, collecting eggs, and caring for chicks, whereas men are more involved in purchasing inputs, marketing birds, transporting products, and making larger production and investment decisions. Because many poultry-management activities are performed within or near the homestead, poultry production provides one of the few agricultural enterprises through which rural women can participate in income-generating activities while continuing to fulfill household responsibilities.

These institutional characteristics imply that variation in local poultry prices is unlikely to affect all household members in the same way. Higher output prices increase both the profitability of poultry production and the opportunity cost of family labor, but the resulting labor adjustments are expected to differ across men and women because they specialize in different production tasks and face different labor-market opportunities. Together, these institutional features provide a natural setting for examining how variation in local poultry prices is transmitted through household-specific shadow wages and household production resources to gender-specific labor allocation under imperfect rural labor markets.

2.3 Local Poultry Markets and Price Variation

Poultry products in Bangladesh are marketed primarily through decentralized village and subdistrict markets that connect smallholder producers with local consumers, traders, and wholesalers (Larive International B.V. and LightCastle Partners Ltd., 2023). Although improvements in transportation infrastructure have

strengthened market integration over time, most rural households continue to purchase inputs and sell poultry products through nearby local markets. Transportation costs, perishability, and the decentralized nature of poultry marketing limit complete price equalization across villages, allowing local supply and demand conditions to remain an important source of price variation for smallholder producers whose marketing activities are concentrated within local trading networks.

Chicken and egg markets also differ in ways that are likely to generate distinct production incentives. Live chickens are commonly marketed through local trading networks and incur higher transport costs because birds are transported live and may experience weight loss during transit. Eggs, by contrast, can be graded, packed, and transported more efficiently, facilitating wider marketing networks and greater spatial arbitrage (Larive International B.V. and LightCastle Partners Ltd., 2023; Sonaiya and Swan, 2004). Consequently, local chicken and egg prices need not move together across villages or over time, creating distinct economic incentives for poultry-producing households.

These market characteristics provide the empirical setting for the analysis that follows. Spatial and temporal variation in village-level chicken and egg prices generates differences in the returns to household poultry production across otherwise similar households. Within the agricultural household framework, these price differences alter the marginal value product of family labor and household production resources. The analysis therefore examines how variation in local poultry prices is transmitted through household-specific shadow wages and household production resources to the labor allocation of male and female household members under imperfect rural labor markets.

3 Conceptual framework

3.1 Agricultural Household Model

The analysis is based on the non-separable agricultural household model, in which production, consumption, and labor allocation are jointly determined when factor markets are imperfect (Janvry and Sadoulet, 1992; Singh et al., 1986). Under complete markets, households maximize farm profits and subsequently choose consumption and labor supply using observed market prices, implying that production and labor decisions are separable. In contrast, when labor markets are incomplete or characterized by transaction costs, production decisions influence household labor allocation through the household-specific opportunity cost of family labor. Consequently, observed market wages no longer provide the relevant measure of the opportunity cost of family labor.

Within this framework, the opportunity cost of family labor is represented by the household-specific shadow wage, defined as the marginal value product of labor in household production (Jacoby, 1993; Skoufias, 1994). Production also generates household-specific shadow income, which represents the production resources available for household consumption after production decisions have been made. Household labor allocation therefore depends on these endogenous shadow values rather than observed market wages and can be expressed as

$$L_{ij} = L(w_{ij}^*, y_i^*), \quad (1)$$

where L_{ij} denotes the labor supplied by individual j in household i , w_{ij}^* is the household-specific shadow wage, and y_i^* denotes household-specific shadow income.

Changes in local poultry prices alter household production incentives and thereby affect both the marginal value product of family labor (the household-specific shadow wage) and household production resources (represented by household-specific shadow income). The empirical analysis therefore examines whether variation in local poultry prices influences household labor allocation through these two channels, corresponding to the substitution and income effects predicted by the agricultural household model. The formal derivation of the model and the comparative statics are presented in Online Appendix 7.

3.2 Local Poultry Prices and Household Labor Allocation

Figure 1 summarizes the mechanism linking variation in local poultry prices to household labor allocation within the agricultural household framework. Local poultry prices influence labor decisions indirectly by changing household production incentives. Higher output prices increase the value of poultry production, thereby raising both the household-specific shadow wage and household-specific shadow income. Under non-separable labor markets, these household-specific shadow values, rather than observed market wages, determine household labor allocation.

The framework predicts two channels through which variation in local poultry prices affects household labor allocation. First, higher output prices increase the household-specific shadow wage, raising the opportunity cost of allocating family labor outside poultry production and encouraging households to reallocate labor toward poultry production through a substitution effect. Second, higher output prices increase household-specific shadow income, generating an income effect that may either reinforce or offset the substitution effect depending on household preferences and labor-market opportunities. Consequently, the overall labor response cannot be inferred from reduced-form price effects alone.

Because men and women specialize in different production tasks and face different labor-market opportunities, identical changes in local poultry prices may generate different changes in their household-specific shadow wages and, consequently, different labor responses. Given women's primary role in poultry husbandry in Bangladesh, local poultry prices are expected to have a larger influence on women's labor allocation, although the magnitude of this difference remains an empirical question.

Following these theoretical predictions, the empirical analysis proceeds in three steps. First, it estimates the reduced-form relationship between local poultry prices and household labor allocation. Second, it recovers household-specific shadow wages and shadow income from the estimated production function. Finally, it examines whether variation in local poultry prices influences the labor supply of male and female household members through changes in household-specific shadow wages and shadow income.

3.3 Empirical Predictions

The conceptual framework yields four empirical predictions.

First, higher local poultry prices are expected to increase the value of household poultry production, thereby raising both household-specific shadow wages and household-specific shadow income.

Second, higher household-specific shadow wages increase the opportunity cost of allocating family labor outside poultry production, generating a substitution effect toward household poultry production. Higher household-specific shadow income generates an income effect that may either reinforce or offset the substitution effect. Consequently, the overall effect of local poultry prices on household labor allocation is theoretically ambiguous and must be determined empirically.

Third, because men and women specialize in different production tasks and face different labor-market opportunities, changes in local poultry prices may generate different labor responses across household members.

Finally, given women's larger role in poultry husbandry in Bangladesh, changes in local poultry prices are expected to have a relatively stronger influence on women's labor allocation than on men's, although the magnitude of this difference remains an empirical question.

4 Data, sample construction, and price measurement

4.1 Data

The empirical analysis draws on the three waves of the Bangladesh Integrated Household Survey (BIHS), a nationally representative panel survey of rural households conducted by the International Food Policy Research Institute (IFPRI) in 2011–12, 2015, and 2018–19 ([Ahmed, 2013, 2016, 2020](#)). The BIHS provides detailed information on agricultural production, livestock ownership, production inputs, individual labor allocation, household consumption, and income sources, making it particularly well suited for analyzing household production and labor decisions within the agricultural household framework.

The survey covers all rural administrative divisions of Bangladesh using a two-stage stratified sampling design. The baseline survey includes 5,503 households drawn from 275 villages. Subsequent survey rounds successfully re-interviewed most original households while following split households to preserve the representativeness of the rural population. Overall panel attrition between the first and third survey rounds was below 7 percent. The geographic distribution of the survey villages is presented in Appendix [Figure A.1](#).

A key strength of the BIHS is its combination of detailed production, labor, and consumption data within a single longitudinal survey. The panel structure provides substantial variation across households, villages, and survey rounds in poultry production, local poultry prices, and labor allocation, while allowing the analysis to control for unobserved time-invariant household characteristics. Unlike most household surveys, the BIHS simultaneously records production inputs, poultry outputs, individual labor activities, and household income, making it possible to estimate household production functions, recover household-specific shadow wages and shadow income, and examine how variation in local poultry prices influences the labor allocation of male and female household members within a unified empirical framework.

The following subsections describe the construction of the estimation sample and the measurement of

village-level poultry prices used in the empirical analysis.

4.2 Estimation Sample

The primary estimation sample is a balanced panel of poultry-producing households observed in all three waves of the Bangladesh Integrated Household Survey (BIHS). A household is classified as a poultry producer if it reported producing chickens, eggs, or both during the survey year. Because household-specific shadow wages and shadow income are recovered from household production functions, the empirical analysis focuses on households engaged in poultry production. Restricting the analysis to households observed in every survey round allows the study to exploit within-household changes over time while controlling for time-invariant household heterogeneity through household fixed effects.

Household production functions are estimated using household-level observations with complete information on poultry output, family labor inputs, production expenditures, and other production inputs. Labor-supply equations are estimated separately for male and female household members aged 15–64 years, corresponding to the working-age population. Household-specific shadow wages and shadow income recovered from the production function are subsequently linked to individual labor-supply decisions, allowing the analysis to examine how household production incentives influence the labor allocation of male and female household members.

[Table A.1](#) summarizes the sample construction. The full BIHS includes 5,503 households in the first survey round, 5,690 in the second, and 6,011 in the third. Restricting the sample to households observed in all three rounds with at least one working-age household member yields a balanced panel of 4,964 households. Among these, 1,614 households reported poultry production in every survey round and constitute the primary estimation sample.

As a robustness check, all analyses are replicated using a broader sample of poultry-producing households observed in at least two survey rounds. Because village-level poultry prices are constructed from household-reported purchase transactions, reliable price measurement requires a sufficient number of transactions within each village. Consequently, in the final estimation household sample observations from two villages are excluded because they do not contain enough purchase transactions to construct reliable local prices. These exclusions affect only a small fraction of the sample (59 household observations per survey round) and do not materially influence the empirical results.

Appendix [Table A.2](#) compares the balanced producer panel with poultry-producing households observed in only two survey rounds. The two groups are broadly comparable across demographic, socioeconomic, production, and labor characteristics. Although households in the balanced panel own slightly more agricultural and homestead land, normalized differences are uniformly small, suggesting that restricting the analysis to the balanced panel does not materially alter the observable composition of the estimation sample.

4.3 Local Poultry Prices

The primary explanatory variables are village-level chicken and egg prices, which proxy the local output prices faced by poultry-producing households. The agricultural household model implies that production

and labor allocation decisions depend on prices prevailing in local markets rather than national or regional averages. In rural Bangladesh, poultry products are marketed primarily through nearby village and subdistrict markets, where transportation costs, market access, and localized supply and demand conditions generate meaningful spatial variation in prices. Accordingly, local poultry prices are measured at the primary sampling unit (PSU) level, which corresponds to villages.

Village-level prices are constructed separately for chickens and eggs using household-reported purchase prices collected during the two-week recall period preceding each survey. For each survey round, the village price is calculated as the average purchase price among purchasing households within the same PSU. When these prices are merged with the household-level data, leave-one-out averages are used so that each household's assigned price excludes its own purchase observation, thereby avoiding a mechanical link between a household's purchasing decision and its assigned local price. When fewer than two purchase observations are available within a village, the corresponding district-level average is used to impute the village price. Such imputations are required for only a small number of villages, primarily in the first survey round, preserving sample coverage while minimizing measurement error arising from sparsely observed local markets.¹

Figure 2 illustrates substantial variation in village-level chicken and egg prices across villages and over time. Chicken prices exhibit greater spatial dispersion than egg prices, consistent with the higher transportation costs and lower spatial market integration associated with live birds. In contrast, egg prices are more tightly distributed, reflecting the greater storability and transportability of eggs. Village-level chicken and egg prices are positively but only moderately correlated, with pairwise correlations of 0.328, 0.189, and 0.111 across the three survey rounds (Appendix Table A.10), indicating that they capture related but distinct dimensions of local poultry markets. These patterns are consistent with evidence that agricultural markets in developing countries exhibit incomplete spatial price transmission because of transportation costs, transaction costs, and other local market frictions (Rashid and Minot, 2010).

To evaluate the validity of the constructed price measures, Figure A.2 compares the purchase-based village prices with independently collected community price data available for the final two BIHS survey rounds². The two measures are positively correlated for both chickens and eggs, indicating that the constructed village prices closely track independently observed local market prices. This agreement provides reassurance that the purchase-based measures capture the local price environment relevant for household production and labor allocation decisions.

4.4 Outcome Measures and Control Variables

The empirical analysis combines individual labor outcomes with household production information to examine how variation in local poultry prices influences labor allocation through household production incentives. The variables used in the analysis can be grouped into four categories: individual labor outcomes,

¹District-level prices are used for 48, 21, and 6 villages for chicken prices and 7, 11, and 1 villages for egg prices in survey rounds 1, 2, and 3, respectively, out of 275 sampled villages.

²Community price data were collected only in the second and third survey rounds through focused group discussion.

household poultry production, production inputs, and household characteristics.

The labor-supply analysis considers both the extensive and intensive margins of labor allocation. The extensive margin is measured by indicators for participation in any income-generating activity during the previous seven days and participation in poultry production. The intensive margin is measured by total weekly hours worked across all income-generating activities and weekly hours devoted to poultry production. Together, these measures capture both participation decisions and adjustments in labor intensity. All labor outcomes are constructed separately for male and female household members aged 15–64 years.

Household poultry production is measured as the annual real value of chicken and egg output reported by the household. Production inputs include annual family labor supplied by women and men, expenditures on feed, medicine, and other variable inputs, the initial value of poultry stock, and the value of productive household assets. These variables represent the principal inputs and outputs of household poultry production and are used to estimate the household production function. Household-specific shadow wages and shadow income recovered from the estimated production function are subsequently incorporated into the labor-supply analysis to examine the channels through which local poultry prices affect labor allocation.

All monetary variables are expressed in constant 2019 prices using the Bangladesh Rural Consumer Price Index and converted to U.S. dollars using the average 2019 exchange rate.³ The empirical specifications additionally control for household demographic characteristics, education, landholdings, productive assets, access to electricity and credit, exposure to crop and livestock shocks, and measures of market access and local infrastructure to account for factors that may jointly influence household production and labor allocation. Detailed definitions and construction of all variables are provided in [Table A.3](#).

5 Empirical Strategy

The empirical strategy combines reduced-form estimation with a production-based approach to examine how variation in local chicken and egg prices affects household labor allocation. The analysis proceeds in four steps. First, it estimates the reduced-form relationship between village-level chicken and egg prices and individual labor outcomes. Second, it estimates a household production function to recover household-specific shadow wages and shadow income. Third, it tests the separability of household production and labor allocation decisions. Finally, it estimates labor-supply equations using the recovered shadow wages and shadow income to distinguish the substitution and income channels through which local poultry prices affect labor allocation. Unless otherwise noted, all specifications include household and survey-round fixed effects, and standard errors are clustered at the village level.

5.1 Reduced-Form Labor Effects

The analysis begins by estimating the reduced-form relationship between local poultry prices and individual labor allocation. These estimates establish whether changes in local poultry prices are associated with labor-market outcomes before examining the economic mechanisms underlying those relationships.

For individual i in household h , village v , and survey round t , the reduced-form specification is

³The conversion uses an exchange rate of 1 BDT = 0.01179 USD.

$$Y_{ihvt} = \beta_{chi} \ln P_{vt}^{chi} + \beta_{egg} \ln P_{vt}^{egg} + X'_{ihvt} \gamma + \lambda_t + \mu_h + \varepsilon_{ihvt}, \quad (2)$$

where Y_{ihvt} denotes one of four labor outcomes: participation in poultry production, weekly poultry hours worked, participation in any income-generating activity, or total weekly hours worked. P_{vt}^{chi} and P_{vt}^{egg} denote village-level chicken and egg prices, respectively. The vector X_{ihvt} includes time-varying individual, household, and village characteristics, λ_t denotes survey-round fixed effects, and μ_h denotes household fixed effects. The models are estimated separately for women and men aged 15–64 years.

Identification comes from within-household changes in village-level poultry prices over time after controlling for household fixed effects, survey-round fixed effects, and observed time-varying covariates. Household fixed effects absorb all time-invariant household characteristics, while survey-round fixed effects capture common macroeconomic shocks affecting all households. Standard errors are clustered at the village level to account for within-village correlation in unobserved market conditions.

The coefficients β_{chi} and β_{egg} capture the overall relationship between local poultry prices and labor allocation. Their signs are theoretically ambiguous because higher poultry prices simultaneously increase the marginal value of family labor, strengthening incentives to work (the substitution effect), and raise household production income, which may reduce labor supply (the income effect). In addition, changes in chicken and egg prices may alter the composition of poultry production and the allocation of labor across economic activities. Consequently, the reduced-form estimates summarize the net effect of local poultry prices on labor allocation but do not separately identify the underlying behavioral channels. The following subsections recover household-specific shadow wages and shadow income from the estimated production function to distinguish these substitution and income effects.

5.2 Estimating the Household Production Function

The agricultural household model implies that labor allocation depends on household-specific shadow wages rather than observed market wages. Because these shadow wages are not directly observed, they are recovered from an estimated household production function following [Jacoby \(1993\)](#), [Skoufias \(1994\)](#), and [Le \(2007\)](#). The estimated production technology also provides the information needed to recover household-specific shadow income, allowing the analysis to distinguish the substitution and income channels through which local poultry prices affect labor allocation.

Household poultry production is modeled using a Cobb–Douglas production function that relates the annual real value of poultry output to family labor and other production inputs. The Cobb–Douglas specification provides a parsimonious representation of household production technology and facilitates recovery of the marginal value products of family labor. Given the three-wave panel and moderate sample size, it also avoids the over-parameterization associated with more flexible functional forms. As a robustness check, I additionally estimate a translog production function, with the results reported in the Online Appendix.

For household h in survey round t , the production function is estimated as

$$\ln Q_{ht} = \alpha + \beta_f \text{IHS} \left(L_{ht}^f \right) + \beta_m \text{IHS} \left(L_{ht}^m \right) + \text{IHS} \left(\mathbf{Z} \right)'_{ht} \boldsymbol{\delta} + \lambda_t + \mu_h + u_{ht}, \quad (3)$$

where Q_{ht} denotes the annual real value of household poultry production, including both chicken and egg output; L_{ht}^f and L_{ht}^m denote annual female and male family labor devoted to poultry production, respectively; and $\mathbf{Z}_{ht}$ includes expenditures on feed, veterinary medicine, and other variable inputs, the initial value of poultry stock, productive household assets, and measures of market access and local infrastructure. Household fixed effects control for time-invariant differences in production technology, managerial ability, and other unobserved household characteristics, while survey-round fixed effects absorb common macroeconomic, technological, and policy shocks. Consequently, the production elasticities are identified from within-household variation in input use over time. Because output is measured as the annual value of production, the estimated marginal products correspond directly to the marginal value products of family labor, allowing household-specific shadow wages to be recovered without an additional output-price adjustment.

Several production inputs contain zero observations, particularly male family labor and expenditures on feed and veterinary services. To retain these observations while preserving a log-like functional form, all continuous production inputs are transformed using the inverse hyperbolic sine (IHS) transformation, which is defined at zero and closely approximates the natural logarithm for larger values (Bellemare and Wichman, 2020). Details of the transformation are provided in Online Appendix 7.

The production function is estimated using the balanced panel of poultry-producing households with household and survey-round fixed effects. Standard errors are clustered at the village level to account for within-village correlation in production shocks and local market conditions. Although the estimated production elasticities are not the primary parameters of interest, they provide the basis for recovering household-specific shadow wages and shadow income, which are subsequently incorporated into the labor-supply analysis to identify the substitution and income effects of local poultry prices.

As with most household production studies, a limitation of this approach is that family labor inputs are self-reported and may therefore be measured with error. As discussed by Skoufias (1994), measurement error in labor inputs can bias the estimated marginal value products from which household-specific shadow wages are recovered. The BIHS does not provide an independent measure of labor inputs that would permit a formal errors-in-variables correction. While the household fixed-effects specification controls for time-invariant household-specific reporting differences, it cannot eliminate time-varying measurement error in labor inputs. Accordingly, the recovered shadow wages should be interpreted subject to this limitation. To assess the sensitivity of the results to the assumed production technology, I also estimate a translog production function. Although this robustness check does not address measurement error, it demonstrates that the principal findings are not sensitive to the assumed production-function specification.

5.3 Recovering Household-Specific Shadow Wages and Shadow Income

Household-specific shadow wages and shadow income are recovered from the estimated household production function following Jacoby (1993), Skoufias (1994), and Le (2007). Under the agricultural household model, the shadow wage equals the marginal value product of family labor. Because the production function

is estimated using the annual real value of poultry output, the estimated marginal products directly measure the household-specific opportunity cost of family labor.

Let $\widehat{Q}_{ht}$ denote the predicted annual value of poultry production obtained from Equation (3). Predicted rather than observed output is used to recover both shadow wages and shadow income, ensuring consistency with the estimated production technology while reducing the influence of transitory production shocks and measurement error (Le, 2007; Makhdum and Kousar, 2021).

For the Cobb–Douglas production function with IHS-transformed labor inputs, the household-specific shadow wage for labor type $j \in \{f, m\}$ is recovered as

$$\widehat{w}_{j,ht}^* = \hat{\beta}_j \frac{\widehat{Q}_{ht}}{\sqrt{(L_{ht}^j)^2 + 1}}, \quad (4)$$

where L_{ht}^j denotes annual family labor supplied by gender j , and $\hat{\beta}_j$ is the estimated coefficient on $\text{IHS}(L_{ht}^j)$. The recovered shadow wage represents the household-specific opportunity cost of allocating an additional unit of family labor to poultry production. As labor inputs increase, Equation (4) converges to the familiar Cobb–Douglas expression for the marginal value product of labor. The derivation is provided in Online Appendix 7.

Household-specific shadow income is defined as

$$\widehat{y}_{ht}^* = \widehat{Q}_{ht} - C_{ht} + Y_{ht}^O, \quad (5)$$

where C_{ht} denotes expenditures on feed, veterinary medicine, and other variable poultry-production inputs, and Y_{ht}^O denotes household income from non-poultry sources. This measure represents the household resources available for consumption after accounting for poultry production costs and income from other activities.

Because household-specific shadow wages and shadow income are generated from the estimated production function, they are potentially endogenous in the labor-supply equations. Following the agricultural household literature (Skoufias, 1994), the following section first tests whether production and labor allocation decisions are separable and then uses an instrumental-variable framework to identify the substitution and income effects of household-specific shadow wages and shadow income on labor allocation.

5.4 Separability Tests

The agricultural household model predicts that production and labor-allocation decisions are separable only when households face complete and competitive labor markets. Under this condition, observed agricultural wages equal the opportunity cost of family labor, so production decisions depend solely on market prices and production technology. When labor markets are imperfect, however, household-specific shadow wages diverge from observed market wages, implying that production and labor-allocation decisions are jointly

determined. This section evaluates whether observed agricultural wages provide an adequate measure of the opportunity cost of family labor. Rejection of the separability assumption supports the use of household-specific shadow wages and shadow income in the subsequent labor-supply analysis.

5.4.1 Jacoby's Test of Separability

Following [Jacoby \(1993\)](#), separability is assessed by comparing the recovered household-specific shadow wage with the observed village-level agricultural wage. Separate regressions are estimated for female and male labor:

$$\ln \widehat{w}_{j,ht}^* = \alpha_j + \beta_j \ln W_{j,vt}^M + \mathbf{Z}_{ht}' \boldsymbol{\gamma} + \lambda_t + \varepsilon_{j,ht}, \quad (6)$$

where $\widehat{w}_{j,ht}^*$ denotes the recovered shadow wage for labor type $j \in \{f, m\}$, $W_{j,vt}^M$ denotes the corresponding village-level agricultural wage, $\mathbf{Z}_{ht}$ includes household characteristics, and λ_t denotes survey-round fixed effects.

Under the null hypothesis of separability,

$$H_0 : \quad \alpha_j = 0 \quad \text{and} \quad \beta_j = 1, \quad (7)$$

so that recovered shadow wages are statistically indistinguishable from observed agricultural wages. The joint hypothesis is evaluated using a Wald test. Rejection of the null indicates that observed agricultural wages do not adequately capture the opportunity cost of family labor, providing evidence of non-separability.

Because observed agricultural wages may be endogenous, Equation (6) is estimated using both ordinary least squares (OLS) and instrumental variables (IV), with the IV specification treated as the preferred model following [Makhdum and Kousar \(2021\)](#). Village-level non-agricultural wages and distance to the nearest market serve as excluded instruments for agricultural wages, reflecting outside employment opportunities and local labor-market integration. Instrument relevance and validity are assessed using conventional first-stage and overidentification tests.

Agricultural wage data are available only in the second and third BIHS survey rounds. Consequently, the Jacoby test is estimated using the balanced panel constructed from these two survey waves.

5.4.2 Le's Generalized Test of Separability

As a complementary assessment, the analysis implements the generalized separability test proposed by [Le \(2010\)](#). Rather than comparing recovered shadow wages with observed agricultural wages, this approach evaluates whether observed agricultural wages alone explain household production decisions.

The estimating equation is

$$\ln \left(\frac{Q_{ht}}{L_{ht}} \right) = \alpha + \sum_{j \in \{f, m\}} \beta_j \ln W_{vt}^{M,j} + \mathbf{Z}_{ht}' \boldsymbol{\gamma} + \lambda_t + \varepsilon_{ht}, \quad (8)$$

where Q_{ht} denotes the annual real value of poultry production, L_{ht} is total family labor devoted to poultry production, $W_{vt}^{M,j}$ denotes the village-level agricultural wage for labor type j , $\mathbf{Z}_{ht}$ includes the remaining production inputs and household characteristics, and λ_t denotes survey-round fixed effects.

Under separability, observed agricultural wages fully capture the opportunity cost of family labor, implying that household characteristics excluded under separability should have no additional explanatory power. The generalized test therefore evaluates the joint null hypothesis

$$H_0 : \beta_f = 1, \quad \beta_m = 1, \quad \text{and} \quad \delta = 0, \quad (9)$$

where δ denotes the coefficients on household characteristics excluded under separability. The null hypothesis is evaluated using a Wald test. Rejection of the joint restrictions provides evidence that household production and labor-allocation decisions are non-separable.

As in the Jacoby test, Equation (8) is estimated using both OLS and IV, with village-level non-agricultural wages and distance to the nearest market serving as excluded instruments for agricultural wages.

Together, the Jacoby and Le tests provide complementary evidence on the validity of the separability assumption. The Jacoby test examines whether recovered shadow wages coincide with observed agricultural wages, whereas the generalized test evaluates whether observed agricultural wages alone explain household production decisions. Rejection of the null hypothesis in both tests indicates that production and labor-allocation decisions are jointly determined, supporting the use of household-specific shadow wages and shadow income in the subsequent instrumental-variable labor-supply analysis.

5.5 Mechanism: Instrumental-Variable Labor-Supply Model

The final stage of the empirical analysis identifies the mechanisms through which local poultry prices affect household labor allocation. Whereas the reduced-form estimates capture the overall effect of local poultry prices on labor outcomes, the instrumental-variable (IV) framework decomposes this relationship into a substitution effect operating through household-specific shadow wages and an income effect operating through household-specific shadow income.

Because household-specific shadow wages and shadow income are recovered from the estimated household production function, they are potentially endogenous to household labor-allocation decisions (Jacoby, 1993; Le, 2007; Skoufias, 1994). Unobserved productivity, managerial ability, and time-varying production shocks may jointly influence production and labor decisions, leading to inconsistent estimates if the recovered variables are treated as exogenous. To address this concern, household-specific shadow wages and shadow income are instrumented using exogenous variation in local poultry prices interacted with predetermined household characteristics.

For individual i in household h , village v , and survey round t , the labor-supply equation is

$$Y_{ihvt} = \alpha + \delta \ln \widehat{w}_{ht}^* + \phi \ln \widehat{y}_{ht}^* + \mathbf{X}_{ihvt}' \boldsymbol{\theta} + \lambda_t + \varepsilon_{ihvt}, \quad (10)$$

where Y_{ihvt} denotes one of four labor outcomes: participation in poultry production, poultry hours worked, participation in any income-generating activity, or total weekly hours worked. The variables $\widehat{w}_{ht}^*$ and $\widehat{y}_{ht}^*$ denote the recovered household-specific shadow wage and shadow income, respectively, $\mathbf{X}_{ihvt}$ contains the individual and household controls included in the reduced-form specification, and λ_t denotes survey-round fixed effects. The model is estimated separately for women and men aged 15–64 years using the corresponding gender-specific household-specific shadow wage together with household-specific shadow income.

The coefficient δ captures the substitution effect of changes in the household-specific opportunity cost of family labor, while ϕ captures the income effect arising from changes in household production resources.

The endogenous variables are instrumented using interactions between village-level chicken and egg prices and predetermined household characteristics. Specifically, the excluded instruments consist of interactions between local poultry prices and (i) the household’s initial poultry stock and (ii) household assets per capita. These interactions exploit the fact that households with larger initial poultry endowments or greater productive assets experience larger changes in the marginal value product of family labor and household production income following changes in local poultry prices. Conditional on household characteristics, village-level poultry prices, and survey-round fixed effects, the interaction terms affect labor allocation only through their effects on household-specific shadow wages and shadow income. Instrument relevance is assessed using conventional first-stage diagnostics, and Hansen’s J statistic is used to evaluate the validity of the overidentifying restrictions.

The baseline specification relates female (male) labor outcomes to the corresponding female (male) household-specific shadow wage together with household-specific shadow income, consistent with the agricultural household model in which labor supply responds to its own opportunity cost while household production resources influence labor allocation through shadow income. Standard errors are clustered at the village level to account for within-village correlation in production shocks and local labor-market conditions. First-stage estimates and identification diagnostics are reported alongside the IV results.

6 Results

This section presents the empirical results in five steps. It begins with descriptive statistics and the estimated production technology, then examines the separability assumption before presenting the reduced-form and instrumental-variable evidence on how local poultry prices affect household labor allocation.

6.1 Descriptive Statistics

Table 1 reports summary statistics for the balanced panel of 1,614 poultry-producing households observed in all three BIHS survey rounds. The estimation sample includes between 2,248 and 2,541 working-age women and between 1,847 and 2,082 working-age men across 260 villages.

The descriptive statistics reveal a pronounced gender specialization in household labor allocation. Poultry production is overwhelmingly managed by women: roughly two-thirds to three-quarters of working-age women participate in poultry production, compared with fewer than four percent of men. Women also devote substantially more time to poultry husbandry, averaging about four hours per week, while men contribute

very little labor to poultry production. By contrast, men are considerably more engaged in other income-generating activities, with labor-force participation exceeding 85 percent and average weekly labor supply approaching full-time employment. These patterns imply that changes in poultry prices are likely to affect women’s labor allocation more strongly than men’s and are consistent with the institutional setting described in Section 2 and motivate the paper’s focus on gender-specific labor responses.

The production statistics reinforce this pattern. The expansion in poultry production also generates substantial within-household variation in recovered shadow wages and shadow income, which is exploited in the subsequent analysis. Throughout the period, female family labor remains the dominant labor input, while male family labor plays only a limited role. Household characteristics also evolve gradually, reflecting improvements in education, household assets, and access to electricity over time.

The summary statistics also indicate substantial variation in local poultry prices, the principal source of identifying variation in the empirical analysis. Real chicken and egg prices differ considerably across villages within each survey round and exhibit meaningful changes over time (see also [Figure 2](#)). Combined with heterogeneity in household production capacity, this variation provides a suitable setting for examining how local poultry prices influence labor allocation through household-specific shadow wages and household production resources.

6.2 Production Function Estimates

[Table 2](#) reports the estimated household poultry production function. Because the production function is used to recover household-specific shadow wages and shadow income, the discussion focuses on the household fixed-effects estimates. The Hausman test rejects the random-effects specification, supporting the use of household fixed effects to account for time-invariant differences in production technology, managerial ability, and other unobserved household characteristics. As discussed in [Section 5.2](#), the Cobb–Douglas specification serves as the baseline, while estimates from the translog specification are reported as a robustness check in [Appendix Table A.4](#).

The estimated production technology highlights the central role of family labor in household poultry production. Female family labor is positively associated ($\beta_f = 0.103$) with poultry production, with a larger estimated coefficient than male family labor ($\beta_m = 0.069$). Because labor inputs enter the production function through the inverse hyperbolic sine transformation, the implied marginal effect varies with the level of labor input. Nevertheless, the larger coefficient indicates that increases in female labor generate larger increases in production than comparable increases in male labor over most of the observed range of the data. This pattern is consistent with the institutional structure of Bangladesh’s smallholder poultry sector, where women undertake most routine poultry management activities while men devote relatively little labor to poultry production. As a result, changes in local poultry prices are expected to generate larger changes in the opportunity cost of female family labor than of male family labor.

Among the remaining production inputs, the initial value of poultry stock is the strongest predictor of production. This result indicates that productive poultry capital amplifies households’ ability to translate favorable price shocks into higher production and household income. Expenditures on feed, veterinary

medicine, and other operating inputs are also positively associated with output, while productive household assets contribute more modestly. By contrast, distance to the nearest town and recent crop or livestock shocks are not statistically associated with poultry production after accounting for household fixed effects and other production inputs.

Overall, the estimated production function is consistent with the production structure of rural poultry enterprises in Bangladesh, where output depends primarily on female family labor, poultry capital, and complementary purchased inputs. These estimates provide the basis for recovering household-specific shadow wages and shadow income, which are used in the subsequent analysis to examine the mechanisms through which local poultry prices affect household labor allocation.

6.3 Household-Specific Shadow Wages and Shadow Income

Table 3 reports summary statistics for the household-specific shadow wages and shadow income recovered from the estimated production function. Panel A presents the measures obtained from the baseline Cobb–Douglas specification, Panel B reports the corresponding measures from the translog specification, and Panel C reports observed village-level agricultural market wages.

The recovered shadow measures exhibit substantial heterogeneity across households. Under the baseline Cobb–Douglas specification, female shadow wages are considerably lower than male shadow wages. This difference reflects the much greater use of female labor in household poultry production. Because the shadow wage equals the marginal value product of family labor, evaluating the marginal product over a substantially larger labor input yields lower hourly shadow wages for women despite their dominant contribution to poultry production. Household-specific shadow income also increases gradually across survey rounds, consistent with the expansion of poultry production documented in the previous subsection.

The wide dispersion in both shadow wages and shadow income indicates that households facing similar local poultry prices nevertheless experience markedly different production incentives because of differences in production capacity, input use, and productive assets. This household-level heterogeneity is central to the empirical analysis, as it generates variation in the opportunity cost of family labor and household production resources beyond that implied by market prices alone.

The shadow measures recovered from the translog specification display patterns similar to those obtained under the Cobb–Douglas specification, indicating that the recovered opportunity costs are not sensitive to the choice of production-function specification. Because the translog specification exhibits weaker identification in the subsequent instrumental-variable analysis, however, the Cobb–Douglas specification is retained as the preferred baseline.

Comparing the recovered shadow wages with observed village-level agricultural wages reveals substantial differences between household-specific opportunity costs and market wages. On average, female shadow wages are lower than prevailing agricultural wages, whereas male shadow wages exceed observed market wages. Moreover, the distribution of household-specific shadow wages is considerably wider than that of observed market wages, reflecting differences in household productivity and production conditions that cannot be captured by a common village-level wage. These descriptive patterns suggest that observed

agricultural wages may not adequately measure the opportunity cost of family labor.

The following subsection formally evaluates this hypothesis using the Jacoby and Le separability tests. Rejection of the separability assumption would imply that household production and labor-allocation decisions are jointly determined, providing empirical support for using household-specific shadow wages and shadow income in the subsequent instrumental-variable labor-supply analysis.

6.4 Separability Tests

The agricultural household model predicts that production and labor-allocation decisions are separable only if observed agricultural wages accurately reflect the opportunity cost of family labor. This prediction is evaluated using the complementary tests proposed by [Jacoby \(1993\)](#) and [Le \(2010\)](#). As discussed in [Section 5.4](#), the instrumental-variable (IV) specifications are treated as the preferred estimates because observed agricultural wages may be endogenous to local labor-market conditions and unobserved household productivity.

[Table 4](#) reports the results of the Jacoby test. Under the preferred IV specification, the joint Wald tests strongly reject the null hypothesis that household-specific shadow wages are statistically equivalent to observed agricultural wages for both female and male labor ($p < 0.001$). The OLS estimates lead to the same conclusion. These results indicate that observed agricultural wages do not adequately measure the opportunity cost of family labor and therefore cannot fully explain household production decisions.

[Table 5](#) presents the results of Le's generalized separability test. Consistent with the Jacoby test, the joint restrictions implied by the separable agricultural household model are strongly rejected under both the IV and OLS specifications ($p < 0.001$). The similarity of the two sets of estimates indicates that the conclusion is robust to accounting for the potential endogeneity of observed agricultural wages.

Taken together, the two separability tests provide strong evidence that production and labor-allocation decisions are jointly determined for poultry-producing households in rural Bangladesh. Household-specific opportunity costs are statistically different from observed agricultural wages, implying that a common market wage does not adequately characterize the labor-allocation decisions of farm households. These findings are consistent with previous evidence from developing-country agriculture ([Makhdom and Kousar, 2021](#); [Skoufias, 1994](#)) and provide empirical support for using household-specific shadow wages and shadow income in the subsequent instrumental-variable labor-supply analysis.

6.5 Reduced-Form Effects of Poultry Prices on Labor Allocation

[Table 6](#) reports the reduced-form relationship between village-level chicken and egg prices and individual labor allocation. These estimates capture the overall effect of local poultry price variation on labor outcomes without distinguishing the mechanisms through which prices influence household behavior. They therefore provide a benchmark for the subsequent analysis, which decomposes the overall price effect into substitution and income effects operating through household-specific shadow wages and shadow income.

Among women, local poultry prices are associated with economically meaningful changes in labor allocation. A one-percent increase in village-level chicken prices is associated with a 0.081 percentage-point increase in

participation in poultry production and a 2.84-hour increase in total weekly labor supply. In contrast, higher egg prices are associated with a 0.171 percentage-point reduction in participation in income-generating activities and a 7.27-hour reduction in total weekly labor supply. Although the individual estimates for poultry hours are imprecisely estimated, the joint tests indicate that chicken and egg prices are jointly associated with both poultry hours ($p = 0.091$) and total weekly labor supply ($p = 0.097$). Taken together, these estimates indicate that fluctuations in local poultry prices lead women to adjust both their participation decisions and labor supply.

The corresponding responses among men are considerably weaker. While higher chicken prices are associated with an increase of approximately 4.85 weekly labor hours and higher egg prices are negatively associated with poultry participation, none of the joint tests rejects the null hypothesis of no overall price effect. Thus, once the combined effects of chicken and egg prices are considered, there is little evidence that local poultry prices systematically affect men's labor allocation.

The stronger responses among women are consistent with the organization of household poultry production in rural Bangladesh, where women provide most routine poultry labor and therefore experience larger changes in the marginal value product and the recovered shadow wage of family labor when local poultry prices change. More broadly, the reduced-form estimates establish that local poultry prices influence household labor allocation but do not reveal whether these responses arise from changes in the opportunity cost of family labor (substitution effects), changes in household production resources (income effects), or both. The next subsection addresses this question by relating labor outcomes directly to household-specific shadow wages and shadow income within the instrumental-variable framework.

6.6 Mechanisms: IV Estimates of Household Labor Supply

The reduced-form estimates establish that local chicken and egg prices are associated with changes in household labor allocation but do not identify the mechanisms underlying these responses. To isolate these mechanisms, [Table 7](#) relates individual labor outcomes directly to recovered household-specific shadow wages and household-specific shadow income. The corresponding first-stage regressions ([Table 8](#)) show how the excluded instruments generate variation in these endogenous variables. Consistent with the production model, interactions between village-level chicken and egg prices and predetermined household productive assets capture differences in households' ability to translate common price shocks into the marginal value product of family labor and the production-profit component of household-specific shadow income. The first-stage estimates indicate that these interactions jointly generate meaningful variation in household-specific shadow wages and shadow income. These patterns are broadly consistent with the production model, whereby households with greater productive capacity experience larger changes in household-specific opportunity costs following changes in local chicken and egg prices.

The IV diagnostics support the identification strategy. The first-stage regressions indicate that the excluded instruments are jointly relevant, with Sanderson–Windmeijer conditional first-stage statistics substantially exceeding conventional thresholds for both endogenous variables. In addition, the second-stage diagnostics reported in [Table 7](#) reject underidentification (Kleibergen–Paap) and fail to reject the validity of the overiden-

tifying restrictions (Hansen's J test). Together, these results indicate that the excluded instruments provide sufficiently strong variation in household-specific shadow wages and household-specific shadow income for the IV estimation.

The IV estimates indicate that household-specific shadow wages primarily affect the intensity of labor devoted to poultry production rather than overall labor supply. Among women, higher household-specific shadow wages increase the probability of participating in poultry production by approximately four percentage points and increase weekly poultry labor by about 1.2 hours, while the estimated effects on overall labor-force participation and total weekly labor supply are small and statistically insignificant. These findings suggest that improvements in household-specific returns increase women's labor devoted to poultry production, although the estimates do not distinguish whether this reflects labor reallocation across activities or a modest expansion in total labor supplied. For men, higher household-specific shadow wages significantly increase weekly poultry labor by approximately 0.4 hours but have no detectable effect on poultry participation or overall labor supply. This pattern is consistent with sectoral labor reallocation, whereby improvements in household-specific returns to poultry production increase labor devoted to poultry activities without expanding total labor supplied. This evidence highlights labor reallocation within the household as an important adjustment mechanism under imperfect labor markets. These findings are consistent with the non-separable agricultural household model, in which changes in household-specific opportunity costs alter the allocation of family labor across competing household activities (Benjamin, 1992; Jacoby, 1993; Le, 2007; Skoufias, 1994).

Household-specific shadow income operates through a different mechanism. Higher household-specific shadow income is associated with greater total weekly labor supply for both women and men, whereas the estimated effects on poultry participation and poultry hours are small and statistically insignificant. The similarity of these responses across women and men suggests that, under the preferred specification, increases in household-specific production resources generate broader adjustments in household labor allocation rather than gender-specific labor-supply responses. Household-specific shadow income combines the production-profit component of household income with exogenous household income and therefore reflects the household's overall economic resources rather than production returns alone. Consequently, the positive association with total labor supply should not be interpreted as a conventional wealth effect. Instead, the results suggest that increases in household-specific shadow income are associated with greater labor supplied across income-generating activities, consistent with higher household-specific returns to production rather than a reduction in labor supply arising from higher income alone.

The estimated magnitudes are economically plausible. Because household-specific shadow wages and household-specific shadow income enter the regressions in logarithmic form while labor outcomes are measured in levels, the estimated coefficients represent changes in labor associated with proportional changes in the recovered shadow measures. Appendix Table A.X reports the implied labor responses associated with a one-standard-deviation increase in household-specific shadow wages and household-specific shadow income. These implied changes are modest relative to average weekly labor supply, indicating economically meaningful but realistic responses.

Overall, the IV estimates identify two complementary channels through which changes in local chicken and egg prices influence household labor allocation under imperfect labor markets. Changes in household-specific shadow wages increase the intensity of labor devoted to poultry production, whereas changes in household-specific shadow income are associated with broader adjustments in total labor supply. These findings explain the reduced-form evidence by showing that changes in local chicken and egg prices influence household labor allocation through household-specific opportunity costs rather than observed market wages. More broadly, the results provide new empirical evidence on how local variation in chicken and egg prices shapes household labor allocation through recovered household-specific shadow wages and household-specific shadow income within a non-separable agricultural household framework, extending the empirical literature on shadow wages and family labor allocation in developing-country agriculture (Beck et al., 2019; Jacoby, 1993).

6.7 Robustness Checks

Several additional analyses demonstrate that the main findings are robust to alternative specifications and sample definitions. First, Appendix Table A.5 reports the corresponding OLS estimates. Relative to the preferred IV specification, the OLS estimates differ substantially in both magnitude and statistical significance, suggesting that failure to account for the endogeneity of household-specific shadow wages and shadow income leads to biased estimates of their effects on labor allocation.

Second, Appendix Table A.7, with the corresponding first-stage results reported in Appendix Table A.8, re-estimates the IV models using the broader sample of poultry-producing households observed in at least two survey rounds. The estimated coefficients remain qualitatively similar in both sign and substantive interpretation, indicating that the main conclusions are not sensitive to restricting the analysis to the balanced panel.

Finally, Appendix Table A.6 reports IV estimates based on shadow wages and shadow income recovered from a translog production function. Although the recovered shadow measures are broadly comparable to those obtained under the baseline Cobb–Douglas specification, the associated IV models exhibit weak first-stage identification, resulting in substantially less precise estimates. Accordingly, the Cobb–Douglas specification is retained as the preferred baseline, and the translog estimates are presented as a robustness exercise rather than an alternative specification.

Overall, these robustness checks indicate that the main findings are not driven by endogeneity, sample selection, or the choice of production-function specification. Together, they reinforce the conclusion that local poultry prices influence household labor allocation through household-specific shadow wages and shadow income.

7 Conclusion

This paper examines how local chicken and egg price fluctuations influence household labor allocation under imperfect labor markets in rural Bangladesh. Although agricultural commodity prices are widely recognized as important determinants of rural household welfare (Lederman and Porto, 2016), comparatively little is known about the mechanisms through which they affect labor allocation when observed market wages do not reflect the opportunity cost of family labor. Building on the non-separable agricultural household model,

this paper combines village-level poultry price variation with household-specific shadow wages and shadow income recovered from an estimated production function to identify these mechanisms.

The analysis yields three main findings. First, both the Jacoby and Le separability tests reject the hypothesis that observed agricultural wages adequately measure the opportunity cost of family labor, supporting the use of household-specific shadow wages to characterize household labor allocation. Second, local poultry price fluctuations primarily affect women's labor allocation, reflecting women's central role in household poultry production. Third, the IV estimates indicate that commodity price shocks operate through two distinct channels: changes in household-specific shadow wages primarily shift labor toward poultry production, whereas changes in household-specific shadow income are associated with broader adjustments in overall labor supply. Together, these findings indicate that household responses to commodity price shocks are governed by endogenous household opportunity costs rather than observed market wages.

These findings extend the agricultural household literature by identifying how local commodity-price variation is transmitted to household labor allocation through household-specific shadow wages and shadow income. While previous studies have emphasized the importance of shadow wages under imperfect labor markets ([Benjamin, 1992](#); [Jacoby, 1993](#); [Le, 2007](#); [Skoufias, 1994](#)), comparatively little empirical evidence has examined how commodity-price shocks are transmitted through these household-specific opportunity costs. By combining spatial variation in village-level poultry prices with recovered shadow values, this paper provides new empirical evidence that commodity-price shocks influence household labor allocation through both substitution effects, captured by changes in household-specific shadow wages, and household resource effects, captured by changes in household-specific shadow income.

The findings also have important implications for agricultural policy. Policies that influence agricultural commodity prices—including investments in livestock value chains, market integration, and producer support programs—are likely to affect rural labor allocation through household production decisions rather than observed wage rates alone. Because women supply most family labor in poultry production, interventions that improve poultry productivity or strengthen producer incentives are also likely to generate disproportionately larger effects on women's economic activities. More broadly, the results underscore the importance of incorporating labor-market imperfections and household production behavior when evaluating the welfare and distributional consequences of agricultural price policies.

The analysis focuses on poultry-producing households in Bangladesh, and the findings may not generalize to other agricultural sectors or institutional settings. Nevertheless, the empirical framework can be readily applied to other agricultural systems in which family labor and imperfect labor markets play an important role. Future research could extend the analysis to other agricultural sectors, examine dynamic labor and investment responses to persistent commodity-price shocks, and incorporate hired labor decisions within a unified household production framework.

This paper highlights the importance of moving beyond observed market wages to household-specific opportunity costs when analyzing labor allocation under imperfect labor markets. Accounting for these endogenous shadow values provides a richer framework for understanding how commodity-price shocks shape household

labor decisions and for evaluating agricultural price policies in developing-country agriculture.

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Figures and Tables

Figure 1: Conceptual Framework: Poultry Prices and Household Labor Allocation

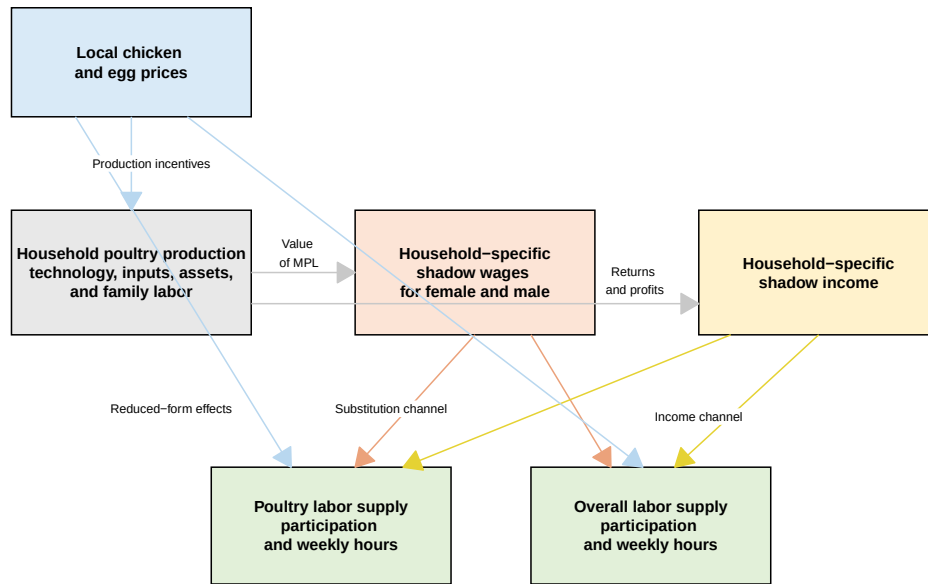


Figure 2: Distribution of real chicken, egg prices over survey rounds

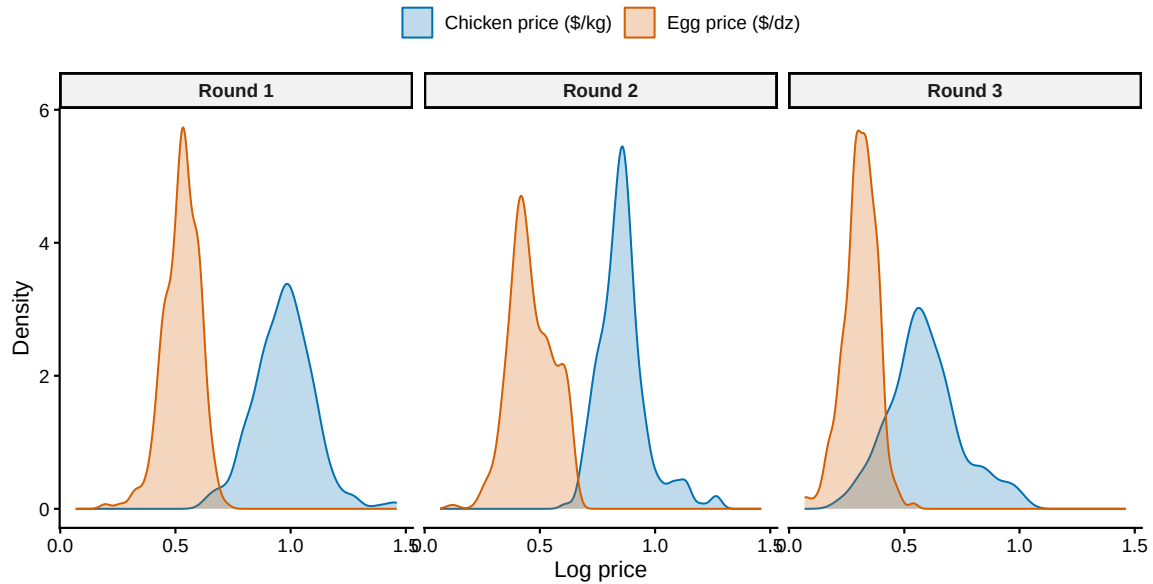


Table 1: Summary statistics for the estimation sample

Variable	Round 1		Round 2		Round 3	
	Mean	SD	Mean	SD	Mean	SD
Panel A: Female characteristics						
<i>Outcomes</i>						
Poultry participation	0.657	0.475	0.703	0.457	0.754	0.431
Poultry weekly hours	3.587	4.014	4.090	4.490	4.105	5.048
Overall participation	0.769	0.422	0.791	0.407	0.829	0.377
Overall weekly hours	10.117	12.183	10.217	12.133	10.465	12.275
<i>Characteristics</i>						
Age	33.492	12.446	34.610	13.220	35.417	13.520
Married	0.780	0.414	0.758	0.429	0.764	0.425
Years of education	4.350	3.904	4.833	4.041	5.201	4.162
Panel B: Male characteristics						
<i>Outcomes</i>						
Poultry participation	0.009	0.096	0.008	0.091	0.036	0.185
Poultry weekly hours	0.067	1.126	0.130	2.116	0.180	1.619
Overall participation	0.881	0.323	0.845	0.362	0.858	0.349
Overall weekly hours	41.304	25.249	37.421	26.697	36.924	26.794
<i>Characteristics</i>						
Age	35.780	13.725	35.723	14.708	35.267	14.691
Married	0.736	0.441	0.682	0.466	0.655	0.475
Years of education	4.685	4.260	5.136	4.386	5.528	4.492
Panel C: Household characteristics						
Household size	4.399	1.517	4.529	1.667	4.574	1.813
Number of children	1.672	1.181	1.560	1.162	1.462	1.177
Number of adult males	1.144	0.721	1.263	0.840	1.290	0.876
Number of adult females	1.393	0.630	1.518	0.737	1.574	0.769
Number of elderly members	0.190	0.429	0.187	0.421	0.248	0.474
Average years of adult education	4.260	3.222	4.684	3.116	5.068	3.116
Household assets (usd/year)	916.530	1573.951	1308.388	1888.137	1539.758	1627.633
Household income w/o poultry (usd/year)	1842.299	3419.803	1989.841	2423.864	2060.921	1956.184
Homestead land (decimals)	11.568	13.415	11.590	12.901	11.976	14.759
Agricultural land (decimals)	102.518	146.613	110.377	168.431	105.119	140.534
Distance to main road (km)	1.989	3.593	1.770	3.822	2.105	3.037
Distance to local market (km)	0.656	0.782	0.508	0.620	0.597	1.393
Distance to district market (km)	1.827	1.724	1.729	1.747	1.912	2.259
Access to electricity	0.490	0.500	0.585	0.493	0.872	0.334
Experienced crop/livestock shock	0.129	0.335	0.074	0.261	0.115	0.319
Household has loan	0.708	0.455	0.739	0.440	0.765	0.424
Panel D: Poultry production						
Value of poultry production (usd/year)	55.489	307.264	60.549	323.745	145.641	1360.339
Initial poultry asset (usd)	29.783	81.336	25.380	70.718	27.999	80.195
Cost for feeding, medicine and other (usd)	11.459	169.890	17.282	300.300	46.826	662.015
Family female labor (hours/year)	228.920	146.697	279.565	157.340	235.990	141.931
Family male labor (hours/year)	6.730	60.685	14.623	96.767	21.929	125.171
Panel E: Local poultry prices						
Chicken price (usd/kg)	2.665	0.353	2.376	0.266	1.825	0.298
Egg price (usd/dz)	1.697	0.126	1.595	0.150	1.366	0.096
Number of female	2248		2450		2541	
Number of male	1847		2039		2082	
Number of households	1614		1614		1614	
Number of local markets (village)	260		260		260	

Notes: The table reports descriptive statistics for the balanced-panel estimation sample consisting of households engaged in poultry production in all three survey rounds. Individual characteristics are calculated for working-age household members aged 15–64, while household characteristics are reported at the household level. Monetary values are expressed in constant 2019 U.S. dollars. Village-level poultry prices represent the average local market prices faced by households in each survey round.

Table 2: Cobb-Douglas production function estimates for household poultry production

	Log of poultry output value		
	OLS (1)	Fixed Effect (2)	Random Effect (3)
Log of family female labor (hours/year)	0.137*** [0.020]	0.103*** [0.024]	0.135*** [0.020]
Log of family male labor (hours/year)	0.068*** [0.010]	0.069*** [0.013]	0.069*** [0.012]
Log of cost for feeding, medicine and others	0.125*** [0.015]	0.133*** [0.019]	0.125*** [0.016]
Log of initial livestock	0.283*** [0.013]	0.219*** [0.017]	0.279*** [0.016]
Log of productive asset	0.049*** [0.007]	0.047*** [0.011]	0.049*** [0.008]
Distance to nearest town (km)	0.000 [0.001]	-0.001 [0.001]	-0.000 [0.001]
Crop and Livestock shock	0.005 [0.045]	0.010 [0.059]	0.005 [0.050]
Constant	1.145*** [0.117]	1.565*** [0.149]	1.167*** [0.127]
Observations	4842	4842	4842
Within R^2	0.271	0.268	0.272
F(Fixed effect)/Wald(Random effect)		61.803	1003.950
F(Fixed effect)/Wald(Random effect): p-value		0.000	0.000
Hausman test χ^2		68.504	

Notes: The production function is estimated using Cobb-Douglas production function. The dependent variable is the log of the annual value of poultry output (2019 USD). Family male labor, family female labor, livestock capital, production expenditures, and productive assets are measured in logarithms. All specifications include survey-round fixed effects. Columns (2) and (3) report household fixed-effects (FE) and random-effects (RE) estimates, with standard errors clustered at the village level reported in brackets. The Hausman test evaluates the null hypothesis that the random-effects estimator is consistent; rejection of the null supports the Kuse of the fixed-effects estimator. The Fixed Effect production-function estimates are subsequently used to recover household-specific shadow wages and shadow income for the labor-supply analysis. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 3: Summary statistics of estimated household shadow wages, shadow income, and market wages

Variable	Round 1			Round 2			Round 3		
	N	Mean	SD	N	Mean	SD	N	Mean	SD
Panel A: Cobb–Douglas production function									
Female shadow wage ($\hat{w}^{*,f}$)	1614	0.046	0.320	1614	0.064	0.578	1614	0.055	0.361
Male shadow wage ($\hat{w}^{*,m}$)	1614	1.996	1.690	1614	2.403	2.099	1614	2.377	2.439
Shadow income per capita ($\hat{y}^*$)	1614	453.759	869.929	1614	468.560	509.710	1614	488.747	527.826
Panel B: Translog production function									
Female shadow wage ($\hat{w}^{*,f}$)	1603	0.022	0.070	1600	0.361	13.465	1603	0.052	0.360
Male shadow wage ($\hat{w}^{*,m}$)	1613	0.982	0.760	1612	1.289	1.017	1614	1.109	1.052
Shadow income per capita ($\hat{y}^*$)	1614	454.295	869.726	1614	469.233	509.237	1614	496.982	524.003
Panel C: Observed agricultural market wages									
Female market wage (W^f)				260	0.385	0.105	260	0.387	0.084
Male market wage (W^m)				260	0.560	0.120	260	0.596	0.148

Notes: The table reports descriptive statistics for the estimated household-specific shadow wages and shadow income recovered from the household production function. Female and male shadow wages are calculated as the marginal value products of family labor using Equation (4), while shadow income is constructed as the predicted value of poultry production net of variable production costs plus household income from non-poultry sources and addition of all exogenous income, as defined in Equation (5). The shadow income is presented on a per-capita basis. All monetary values are expressed in constant 2019 U.S. dollars.

Table 4: [Jacoby \(1993\)](#) test of shadow wage equality

	$\hat{w}^{*,f}$		$\hat{w}^{*,m}$	
	OLS (1)	IV (2)	OLS (3)	IV (4)
W^f	0.011 [0.067]	0.143 [0.220]		
W^m			-0.368 [0.527]	-0.805 [1.163]
Constant	0.059*** [0.022]	0.010 [0.076]	2.605*** [0.305]	2.845*** [0.648]
Year FE	x	x	x	x
Observations	3228	3228	3228	3228
Jacoby test: F-stat/ χ^2	251.644	575.278	292.614	593.586
P-value	0.000	0.000	0.000	0.000

Notes: The table reports [Jacoby \(1993\)](#) separability test, which compares household-specific shadow wages with observed market wages. The dependent variables are the estimated female and male shadow wages. The test evaluates the joint null hypothesis that the intercept equals zero and the coefficient on the corresponding market wage equals one. Rejection of the null implies that observed market wages do not fully represent the opportunity cost of family labor, providing evidence against separability. Standard errors are clustered at the village level. Columns (2) and (4) report IV estimates.

Table 5: Le (2010) generalized test of separability

	$\ln(PQ/L)$	
	OLS (1)	IV (2)
$\ln W^m$	-0.011 [0.123]	0.521 [0.537]
$\ln W^f$	0.026 [0.097]	-0.533 [0.529]
Log of family female labor (hours/year)	-0.424*** [0.039]	-0.424*** [0.039]
Log of family male labor (hours/year)	-0.055*** [0.013]	-0.051*** [0.014]
Log of cost for feeding, medicine and others	0.141*** [0.018]	0.132*** [0.020]
Log of initial livestock	0.221*** [0.019]	0.223*** [0.019]
Log of productive asset	0.039*** [0.009]	0.040*** [0.010]
Distance to nearest town (km)	-0.001 [0.001]	-0.001 [0.001]
Crop and Livestock shock	0.029 [0.056]	0.048 [0.061]
Number of adult male	0.008 [0.023]	0.013 [0.024]
Number of adult female	-0.035 [0.022]	-0.034 [0.023]
Dependency ratio	-0.021 [0.029]	-0.016 [0.030]
Average year of education for adults	0.012** [0.006]	0.011* [0.006]
Year FE	x	x
Observations	3228	3228
Le generalized test:		
F-stat/ χ^2	52.374	99.638
P-value	0.000	0.000

Notes: The dependent variable is the log of the value of poultry output per unit of labor input, $\ln(PQ/L)$. Column (1) reports ordinary least squares (OLS) estimates, while column (2) reports instrumental variables (IV) estimates, which constitute the preferred specification. Both models include year fixed effects and production controls. Standard errors clustered at the village level are reported in brackets. The Le generalized test evaluates the joint null hypothesis that the coefficients on the male and female market wages equal one and that household demographic characteristics contain no additional information for production decisions beyond that captured by market wages, observed labor inputs, and other production inputs. Rejection of the null indicates that production and household labor allocation decisions are jointly determined, providing evidence against separability. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6: Reduced-form effects of local poultry prices on weekly labor supply

	Poultry labor supply		Overall labor supply	
	Participation (1)	Weekly hours (2)	Participation (3)	Weekly hours (4)
Panel A: Female				
$\ln P^{chi}$	0.081* [0.048]	-0.988** [0.463]	0.081** [0.037]	2.836** [1.391]
$\ln P^{egg}$	-0.055 [0.076]	-0.299 [0.723]	-0.171*** [0.064]	-7.270*** [2.345]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	7239	7239	7239	7239
Outcome mean	0.707	3.939	0.797	10.273
<i>p-value: Total price effect</i> $\ln P^{chi} + \ln P^{egg} = 0$	0.764	0.091	0.202	0.097
Panel B: Male				
$\ln P^{chi}$	0.030* [0.017]	0.133 [0.136]	0.025 [0.032]	4.846* [2.871]
$\ln P^{egg}$	-0.047* [0.024]	-0.060 [0.297]	-0.013 [0.052]	0.760 [5.170]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	5968	5968	5968	5968
Outcome mean	0.018	0.128	0.861	38.449
<i>p-value: Total price effect</i> $\ln P^{chi} + \ln P^{egg} = 0$	0.533	0.821	0.817	0.310

Notes: The sample consists of working-age individuals (15–64 years) from the balanced panel of poultry-producing households observed in all three BIHS survey rounds. The dependent variables are indicators for participation in poultry production and any income-generating activity, and the corresponding weekly hours worked. The key explanatory variables are the logarithms of village-level chicken and egg prices. All specifications are estimated by ordinary least squares and include the full set of individual, household, and village control variables described in Appendix Table A.3, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 7: IV estimated of the effects of household-specific shadow wages and shadow income on weekly labor supply

	Poultry labor supply		Overall labor supply	
	Participation (1)	Weekly hours (2)	Participation (3)	Weekly hours (4)
Panel A: Female				
$\ln \widehat{w}^{*,f}$	0.040*	1.161***	-0.005	0.692
	[0.021]	[0.248]	[0.020]	[0.673]
$\ln \widehat{y}^*$	-0.005	0.179	-0.009	1.446*
	[0.024]	[0.254]	[0.022]	[0.786]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	7239	7239	7239	7239
Outcome mean	0.707	3.939	0.797	10.273
<i>Instrument diagnostics:</i>				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	11.114	11.114	11.114	11.114
Hansen J test: p-value	0.403	0.554	0.405	0.912
Panel B: Male				
$\ln \widehat{w}^{*,m}$	0.015	0.422***	-0.001	0.323
	[0.009]	[0.159]	[0.016]	[1.590]
$\ln \widehat{y}^*$	-0.007	-0.141	0.009	6.383***
	[0.011]	[0.241]	[0.031]	[2.327]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	5968	5968	5968	5968
Outcome mean	0.018	0.128	0.861	38.449
<i>Instrument diagnostics:</i>				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	14.549	14.549	14.549	14.549
Hansen J test: p-value	0.501	0.773	0.333	0.182

Notes: The table reports IV estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply for working-age individuals (15–64 years) from the balanced panel of poultry-producing households observed in all three BIHS survey rounds. Panel A reports estimates for women and Panel B reports estimates for men. The endogenous regressors are the logarithms of the household-specific female and male shadow wages and the inverse hyperbolic sine of household-specific shadow income per capita. These endogenous regressors are instrumented using interactions between the logarithms of village-level chicken and egg prices and predetermined household characteristics, namely poultry capital at the beginning of the production year and household assets per capita. All specifications include the full set of individual, household, and village control variables described in Appendix Table A.3, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. The Kleibergen–Paap LM statistic tests the null hypothesis that the model is underidentified, the Kleibergen–Paap rk Wald F statistic assesses the strength of the excluded instruments under clustered standard errors, and Hansen’s J statistic tests the validity of the overidentifying restrictions. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

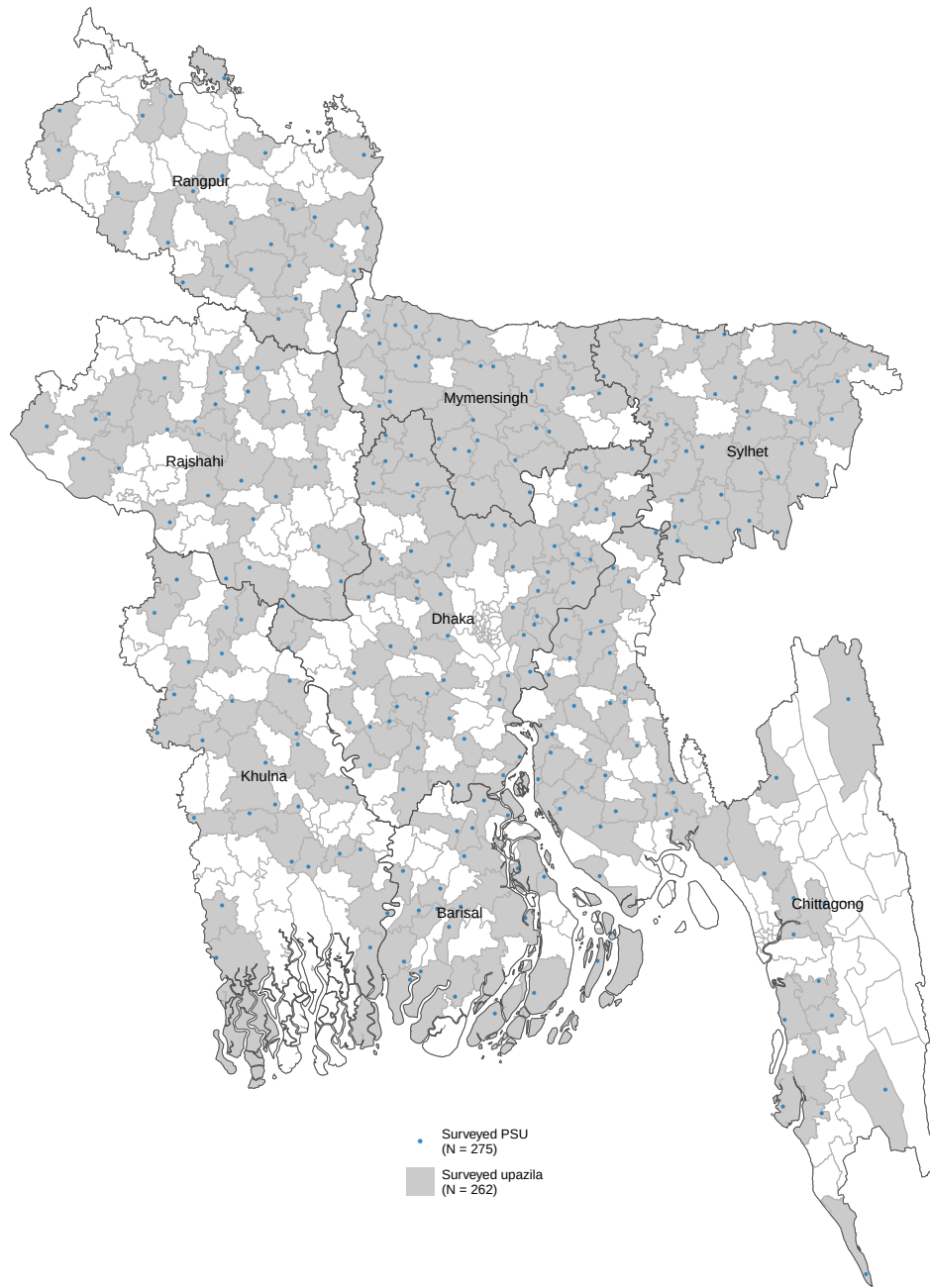
Table 8: First-stage relationships between excluded instruments and endogenous variables

	Female		Male	
	$\ln \widehat{w}^{*,f}$ (1)	$\ln \widehat{y}^*$ (2)	$\ln \widehat{w}^{*,m}$ (3)	$\ln \widehat{y}^*$ (4)
Log of initial poultry capital $\times \ln P^{chi}$	0.234*** [0.087]	0.055 [0.142]	0.260** [0.125]	-0.061 [0.177]
Log of initial poultry capital $\times \ln P^{egg}$	-0.064 [0.160]	-0.233 [0.272]	-0.064 [0.229]	-0.018 [0.331]
Log of per capita asset $\times \ln P^{chi}$	0.008 [0.100]	-0.043 [0.092]	0.105 [0.140]	0.019 [0.116]
Log of per capita asset $\times \ln P^{egg}$	0.049 [0.179]	0.540*** [0.184]	-0.090 [0.252]	0.318 [0.212]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	7239	7239	5968	5968
Partial R^2	0.054	0.023	0.031	0.014
Sanderson–Windmeijer conditional F	53.901	22.885	25.393	19.998

Notes: This table reports the first-stage regressions for the IV labor-supply models. Columns (1) and (2) present the first-stage equations for the female sample, and columns (3) and (4) present the corresponding equations for the male sample. The endogenous variables are household-specific shadow wages and household-specific shadow income. The excluded instruments are interactions between the logarithm of village-level poultry prices (chicken and eggs) and predetermined household productive assets, including initial poultry capital and household assets per capita. All specifications include the full set of individual, household, and community controls and survey-round fixed effects. Standard errors clustered at the village level are reported in brackets. Partial R^2 and Sanderson–Windmeijer (SW) conditional first-stage F statistics are reported for each endogenous variable. Kleibergen–Paap and Hansen’s J statistics are reported with the corresponding second-stage estimates in Table 7. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

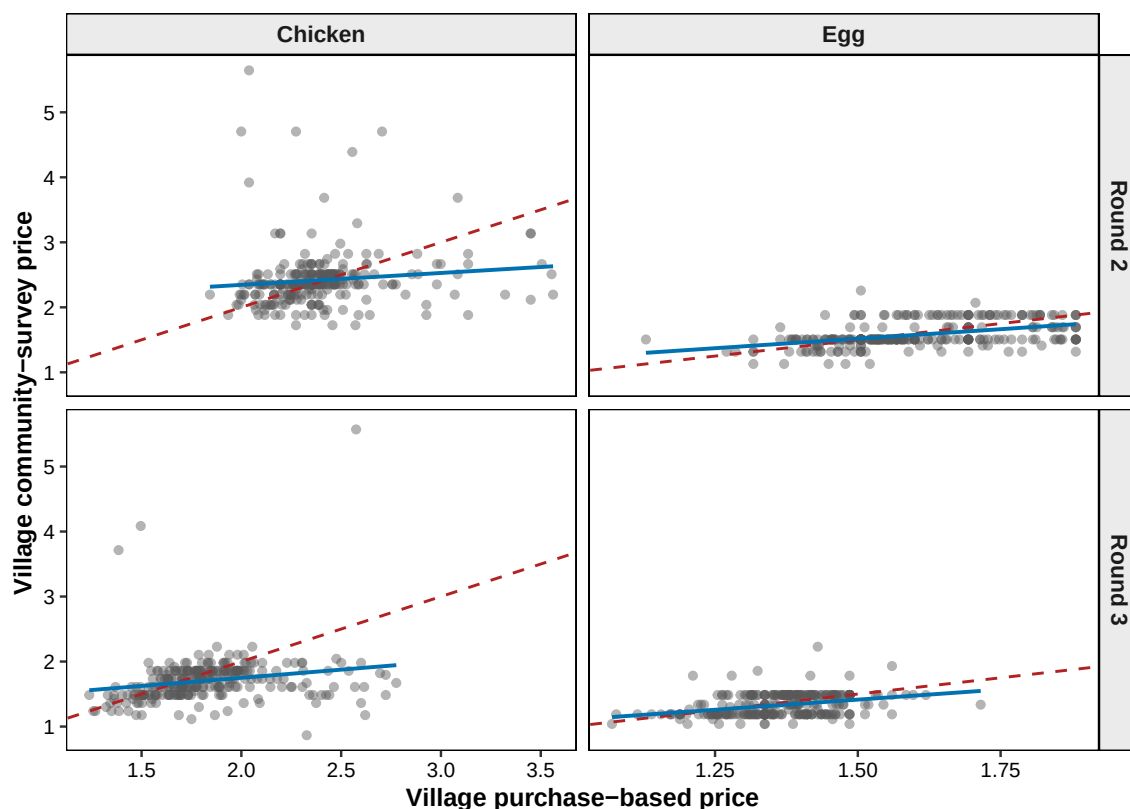
Appendix A. Figures and Tables

Figure A.1: Survey area of Bangladesh Integrated Household Survey



Notes: The production-function estimations use household observations. Labor supply estimations use individual observations for working-age household members (15–64 years). The primary estimation sample consists of the balanced panel of poultry-producing households observed in all three survey rounds. Robustness analyses additionally use poultry-producing households observed in at least two survey rounds.

Figure A.2: Comparison of village-level poultry prices constructed from household purchases and community surveys



Notes: Each point represents a village in a survey round. The dashed red line denotes the 45-degree line, while the blue solid line shows the linear fit. The close correspondence between the two measures indicates that purchase-based village prices closely track independently collected community survey prices

Table A.1: Sample selection for the empirical analysis

Sample	Household observations				Individual observations (15–64)			
	Round 1	Round 2	Round 3	Total	Round 1	Round 2	Round 3	Total
All surveyed households	5503	5690	6011	17204	13404	14104	14215	41723
Households observed in at least two rounds	5341	5670	5405	16416	12714	14039	12946	39699
Balanced household panel (3 rounds)	4964	4964	4964	14892	11548	11992	12017	35557
Poultry-producing households (all)	3497	3463	3717	10677	8813	9293	9796	27902
Poultry-producing households observed in at least two rounds (robustness sample)	2787	2977	2794	8558	7045	8149	7737	22931
Balanced poultry-producing panel (main estimation sample)	1614	1614	1614	4842	4095	4489	4623	13207

Notes: Household observations are used to estimate the production function. Individual observations include working-age household members (15–64 years) used in the labor supply analysis. The primary estimation sample consists of the balanced panel of poultry-producing households observed in all three survey rounds. Robustness analyses additionally use poultry-producing households observed in at least two survey rounds. Observations from two villages are excluded from both producer samples because village-level poultry prices could not be reliably constructed owing to insufficient household purchase observations.

Table A.2: Comparison of the main estimation sample and excluded poultry-producing households

Variable	Round 1			Round 2			Round 3		
	Exc. sample	Est. sample	Norm. Diff.	Exc. sample	Est. sample	Norm. Diff.	Exc. sample	Est. sample	Norm. Diff.
Panel A: Female characteristics									
<i>Outcomes</i>									
Poultry participation	0.638	0.657	0.041	0.684	0.703	0.040	0.759	0.754	-0.010
Poultry weekly hours	3.204	3.587	0.101	3.642	4.090	0.102	4.304	4.105	-0.033
Overall participation	0.747	0.769	0.050	0.784	0.791	0.018	0.845	0.829	-0.045
Overall weekly hours	9.782	10.117	0.026	9.801	10.217	0.034	11.379	10.465	-0.068
<i>Characteristics</i>									
Age	33.967	33.492	-0.037	34.756	34.610	-0.011	34.799	35.417	0.046
Married	0.774	0.780	0.016	0.771	0.758	-0.033	0.756	0.764	0.018
Years of education	3.929	4.350	0.110	4.446	4.833	0.097	4.897	5.201	0.074
Panel B: Male characteristics									
<i>Outcomes</i>									
Poultry participation	0.008	0.009	0.010	0.010	0.008	-0.022	0.038	0.036	-0.012
Poultry weekly hours	0.048	0.067	0.020	0.104	0.130	0.014	0.164	0.180	0.010
Overall participation	0.875	0.881	0.019	0.858	0.845	-0.037	0.877	0.858	-0.056
Overall weekly hours	40.120	41.304	0.048	39.121	37.421	-0.064	36.797	36.924	0.005
<i>Characteristics</i>									
Age	34.560	35.780	0.089	34.829	35.723	0.062	35.605	35.267	-0.023
Married	0.702	0.736	0.077	0.695	0.682	-0.027	0.689	0.655	-0.072
Years of education	4.410	4.685	0.065	4.799	5.136	0.078	5.114	5.528	0.094
Panel C: Household Characteristics									
Household size	4.299	4.399	0.063	4.478	4.529	0.029	4.442	4.574	0.077
Number of children	1.548	1.672	0.106	1.540	1.560	0.017	1.539	1.462	-0.065
Number of adult males	1.129	1.144	0.020	1.190	1.263	0.085	1.164	1.290	0.148
Number of adult females	1.386	1.393	0.010	1.495	1.518	0.031	1.475	1.574	0.133
Number of elderly members	0.236	0.190	-0.102	0.253	0.187	-0.144	0.264	0.248	-0.032
Average years of adult education	3.929	4.260	0.103	4.347	4.684	0.108	4.748	5.068	0.103
Household assets	860.224	916.530	0.036	1154.353	1308.388	0.091	1357.097	1539.758	0.110
Homestead land (decimals)	10.623	11.568	0.074	10.294	11.590	0.106	10.181	11.976	0.142
Agricultural land (decimals)	86.164	102.518	0.106	88.271	110.377	0.135	76.990	105.119	0.224
Distance to main road (km)	2.003	1.989	-0.004	2.232	1.770	-0.095	2.287	2.105	-0.049
Distance to local market (km)	0.621	0.656	0.049	0.484	0.508	0.040	0.751	0.597	-0.046
Distance to district market (km)	1.705	1.827	0.071	1.695	1.729	0.020	1.872	1.912	0.017
Experienced crop/livestock shock	0.116	0.129	0.039	0.067	0.074	0.024	0.098	0.115	0.053
Number of female	1626	2248		2038	2450		1741	2541	
Number of male	1324	1847		1622	2039		1373	2082	
Number of households	1173	1614		1363	1614		1180	1614	

Notes: The estimation sample (Est. sample) consists of poultry-producing households observed in all three survey rounds (balanced panel). The excluded sample (Exc. sample) consists of poultry-producing households observed in exactly two survey rounds. Individual characteristics are reported for working-age household members (15–64 years), while household characteristics are reported at the household level. Normalized differences are calculated as the difference in group means divided by the square root of the average of the two group variances. Absolute normalized differences below 0.10 generally indicate negligible differences between the two groups.

Table A.3: Definitions and measurement of variables used in the empirical analysis

Variable	Definition	Unit or construction
Panel A: Labor outcomes		
Poultry participation	Indicator equal to one if the individual worked in poultry production during the previous seven days, and zero otherwise.	Indicator (0/1)
Poultry weekly hours	Total hours devoted to poultry production during the previous seven days.	Hours per week
Overall participation	Indicator equal to one if the individual participated in any income-generating activity during the previous seven days, including salaried employment, wage work, self-employment, apprenticeship, or work in a family enterprise.	Indicator (0/1)
Overall weekly hours	Total hours devoted to all income-generating activities during the previous seven days.	Hours per week
Panel B: Individual characteristics		
Age	Age of the individual at the time of the survey.	Years
Married	Indicator equal to one if the individual is currently married, and zero otherwise.	Indicator (0/1)
Years of education	Number of completed years of formal schooling.	Years
Panel C: Household characteristics		
Household size	Total number of individuals residing in the household.	Number of persons
Number of children	Number of household members younger than 15 years.	Number of persons
Number of adult males	Number of male household members aged 15–64 years.	Number of persons
Number of adult females	Number of female household members aged 15–64 years.	Number of persons
Number of elderly members	Number of household members aged 65 years or older.	Number of persons
Average years of adult education	Average completed years of schooling among working-age household members aged 15–64 years.	Years
Household assets	Total value of household productive and other included assets.	Constant 2019 USD
Household income excluding poultry	Annual household income from sources other than poultry production.	Constant 2019 USD per year
Homestead land	Total area of land used for the household dwelling and surrounding homestead.	Decimals
Agricultural land	Total area of agricultural land operated or owned by the household, according to the measure used in the empirical analysis.	Decimals
Distance to main road	Distance from the household or village to the nearest main road.	Kilometers
Distance to local market	Distance from the household or village to the nearest local market.	Kilometers
Distance to district market	Distance from the household or village to the relevant district-level market.	Kilometers
Access to electricity	Indicator equal to one if the household has access to electricity, and zero otherwise.	Indicator (0/1)
Experienced crop or livestock shock	Indicator equal to one if the household experienced a crop- or livestock-related production shock during the survey reference period.	Indicator (0/1)
Household has a loan	Indicator equal to one if the household reported an outstanding loan or access to credit during the survey period.	Indicator (0/1)
Panel D: Poultry production variables		
Value of poultry production	Total annual real value of household poultry output, including chickens and eggs produced during the survey year.	Constant 2019 USD per year
Initial poultry assets	Value of poultry stock held at the beginning of the production period.	Constant 2019 USD
Feed, medicine, and other production costs	Annual expenditures on poultry feed, veterinary medicine, and other variable poultry-production inputs.	Constant 2019 USD per year
Female family labor	Total annual hours supplied by female household members to poultry production.	Hours per year
Male family labor	Total annual hours supplied by male household members to poultry production.	Hours per year
Panel E: Local poultry prices		
Chicken price	Leave-one-out village-level average chicken unit value constructed from household-reported purchase transactions during the two-week recall period. District-round averages are used when fewer than two village purchase observations are available.	Constant 2019 USD per kilogram
Egg price	Leave-one-out village-level average egg unit value constructed from household-reported purchase transactions during the two-week recall period. District-round averages are used when fewer than two village purchase observations are available.	Constant 2019 USD per dozen
Panel F: Estimated mechanism variables		
Female shadow wage	Marginal value product of female family labor recovered from the estimated household poultry-production function.	Constant 2019 USD per hour
Male shadow wage	Marginal value product of male family labor recovered from the estimated household poultry-production function.	Constant 2019 USD per hour
Household-specific shadow income PC	Predicted poultry output value net of variable input costs and imputed family-labor costs, plus household income from external sources; divided by household size where the per-capita measure is used.	Constant 2019 USD per capita

Table A.4: Translog production function estimation for household poultry production

	Log of poultry output value		
	OLS (1)	Fixed Effect (2)	Random Effect (3)
Log of family female labor (hours/year)	0.049 [0.093]	0.022 [0.110]	0.047 [0.094]
Log of family male labor (hours/year)	0.101* [0.061]	0.071 [0.076]	0.100 [0.062]
Log of feed, medicine, and other costs (2019 USD)	-0.113* [0.061]	-0.112 [0.071]	-0.113* [0.063]
Log of initial poultry assets (2019 USD)	0.164** [0.065]	0.172** [0.078]	0.163*** [0.062]
Log of productive assets (2019 USD)	0.090** [0.045]	0.050 [0.065]	0.088* [0.049]
0.5 × [Log family female labor] ²	0.039** [0.017]	0.028 [0.021]	0.039** [0.019]
0.5 × [Log family male labor] ²	-0.005 [0.014]	-0.001 [0.018]	-0.005 [0.015]
0.5 × [Log feed, medicine, and other costs] ²	0.121*** [0.012]	0.131*** [0.015]	0.121*** [0.013]
0.5 × [Log initial poultry assets] ²	0.101*** [0.013]	0.068*** [0.022]	0.099*** [0.017]
0.5 × [Log productive assets] ²	-0.010* [0.006]	-0.015 [0.010]	-0.010 [0.006]
Log family female labor × Log family male labor	-0.008 [0.009]	0.000 [0.011]	-0.007 [0.009]
Log family female labor × Log costs	0.020** [0.010]	0.020* [0.011]	0.020** [0.010]
Log family female labor × Log initial poultry assets	-0.024** [0.012]	-0.022 [0.015]	-0.024* [0.013]
Log family female labor × Log productive assets	-0.011 [0.007]	-0.007 [0.009]	-0.011 [0.007]
Log family male labor × Log feed, medicine, and other costs	0.007 [0.006]	0.005 [0.007]	0.007 [0.006]
Log family male labor × Log initial poultry assets	-0.010 [0.008]	-0.012 [0.010]	-0.010 [0.008]
Log family male labor × Log productive assets	0.003 [0.004]	-0.000 [0.005]	0.003 [0.004]
Log costs × Log initial assets	-0.042*** [0.008]	-0.050*** [0.010]	-0.043*** [0.008]
Log costs × Log productive assets	-0.003 [0.005]	0.004 [0.005]	-0.003 [0.005]
Log initial poultry assets × Log productive assets	0.017*** [0.005]	0.024*** [0.006]	0.017*** [0.005]
Distance to nearest town (km)	-0.000 [0.001]	-0.001 [0.001]	-0.000 [0.001]
Crop and livestock shock	0.009 [0.043]	0.025 [0.055]	0.010 [0.047]
Constant	1.331*** [0.307]	1.823*** [0.387]	1.355*** [0.311]
Observations	4842	4842	4842
Adj. R^2	0.324	0.274	.
F(Fixed effect)/Wald(Random effect)		49.359	2194.772
F(Fixed effect)/Wald(Random effect): p-value		0.000	0.000
Hausman test χ^2		88.144	

Notes: The production function is estimated using Translog production function. The dependent variable is the log of the annual value of poultry output (2019 USD). Family male labor, family female labor, livestock capital, production expenditures, and productive assets are measured in logarithms. All specifications include survey-round fixed effects. Columns (2) and (3) report household fixed-effects (FE) and random-effects (RE) estimates, with standard errors clustered at the village level reported in brackets. The Hausman test evaluates the null hypothesis that the random-effects estimator is consistent; rejection of the null supports the use of the fixed-effects estimator. The Fixed Effect production-function estimates are subsequently used to recover household-specific shadow wages and shadow income for the labor-supply analysis. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.5: OLS estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply

	Poultry labor supply		Overall labor supply	
	Participation	Weekly hours	Participation	Weekly hours
	(1)	(2)	(3)	(4)
Panel A: Female				
$\ln \hat{w}^{*,f}$	-0.019*** [0.005]	-0.310*** [0.067]	-0.016*** [0.005]	-0.214 [0.184]
$\ln \hat{y}^*$	-0.002 [0.003]	-0.159*** [0.061]	0.002 [0.003]	0.095 [0.113]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	7239	7239	7239	7239
Outcome mean	0.707	3.939	0.797	10.273
Panel B: Male				
$\ln \hat{w}^{*,m}$	-0.017*** [0.002]	-0.114*** [0.032]	-0.010*** [0.003]	-0.140 [0.255]
$\ln \hat{y}^*$	-0.010*** [0.004]	-0.231*** [0.080]	0.020*** [0.004]	3.004*** [0.406]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	5968	5968	5968	5968
Outcome mean	0.018	0.128	0.861	38.449

Notes: The table reports ordinary least squares (OLS) estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply for working-age individuals (15–64 years) from the balanced panel of poultry-producing households observed in all three BIHS survey rounds. Panel A reports estimates for women and Panel B reports estimates for men. The regressors of interest are the logarithms of the household-specific female and male shadow wages and the inverse hyperbolic sine of household-specific shadow income per capita. All specifications include the full set of individual, household, and village control variables described in Appendix Table A.3, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.6: IV labor-supply estimates using shadow wages and shadow income recovered from the translog production function

	Poultry labor supply		Overall labor supply	
	Participation (1)	Weekly hours (2)	Participation (3)	Weekly hours (4)
Panel A: Female				
$\ln \hat{w}^{*,f}$	0.104 [0.064]	3.117*** [0.994]	-0.011 [0.053]	1.628 [1.932]
$\ln \hat{y}^*$	-0.001 [0.026]	0.252 [0.339]	-0.007 [0.022]	1.519* [0.782]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	7186	7186	7186	7186
Outcome mean	0.707	3.940	0.798	10.300
<i>Instrument diagnostics:</i>				
Kleibergen-Paap LM test: p-value	0.002	0.002	0.002	0.002
Kleibergen-Paap rk Wald F statistic	4.152	4.152	4.152	4.152
Hansen J test: p-value	0.388	0.738	0.444	0.905
Panel B: Male				
$\ln \hat{w}^{*,m}$	-0.056* [0.031]	-1.251* [0.708]	0.041 [0.055]	4.557 [5.134]
$\ln \hat{y}^*$	0.010 [0.015]	0.279 [0.313]	-0.003 [0.038]	5.176* [3.003]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	5962	5962	5962	5962
Outcome mean	0.018	0.125	0.861	38.452
<i>Instrument diagnostics:</i>				
Kleibergen-Paap LM test: p-value	0.083	0.083	0.083	0.083
Kleibergen-Paap rk Wald F statistic	1.773	1.773	1.773	1.773
Hansen J test: p-value	0.914	0.593	0.436	0.370

Notes: The table reports IV estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply for working-age individuals (15–64 years) from the balanced panel of poultry-producing households observed in all three BIHS survey rounds. Panel A reports estimates for women and Panel B reports estimates for men. The endogenous regressors are the logarithms of the household-specific female and male shadow wages and the inverse hyperbolic sine of household-specific shadow income per capita, where the shadow variables are recovered from the translog production function. These endogenous regressors are instrumented using interactions between the logarithms of village-level chicken and egg prices and predetermined household characteristics, namely poultry capital at the beginning of the production year and household assets per capita. All specifications include the full set of individual, household, and village control variables described in Appendix Table A.3, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. The Kleibergen–Paap LM statistic tests the null hypothesis that the model is underidentified, the Kleibergen–Paap rk Wald F statistic assesses the strength of the excluded instruments under clustered standard errors, and Hansen’s J statistic tests the validity of the overidentifying restrictions. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.7: Robustness check: IV labor-supply estimates using the broader sample of poultry-producing households observed in at least two survey rounds

	Poultry labor supply		Overall labor supply	
	Participation (1)	Weekly hours (2)	Participation (3)	Weekly hours (4)
Panel A: Female				
$\ln \hat{w}^{*,f}$	0.045*** [0.016]	1.223*** [0.202]	0.007 [0.014]	1.196** [0.514]
$\ln \hat{y}^*$	-0.029 [0.020]	-0.133 [0.215]	-0.030 [0.019]	0.224 [0.719]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	12644	12644	12644	12644
Outcome mean	0.702	3.847	0.795	10.286
<i>Instrument diagnostics:</i>				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	19.214	19.214	19.214	19.214
Hansen J test: p-value	0.755	0.233	0.472	0.925
Panel B: Male				
$\ln \hat{w}^{*,m}$	0.017** [0.007]	0.403*** [0.108]	0.012 [0.012]	1.038 [0.986]
$\ln \hat{y}^*$	-0.006 [0.008]	-0.107 [0.158]	0.035 [0.025]	5.986*** [1.929]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	10287	10287	10287	10287
Outcome mean	0.018	0.119	0.864	38.550
<i>Instrument diagnostics:</i>				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	22.605	22.605	22.605	22.605
Hansen J test: p-value	0.542	0.116	0.402	0.087

Notes: The table reports IV estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply for working-age individuals (15–64 years) from the broader sample of poultry-producing households observed in at least two BIHS survey rounds. Panel A reports estimates for women and Panel B reports estimates for men. The endogenous regressors are the logarithms of the household-specific female and male shadow wages and the inverse hyperbolic sine of household-specific shadow income per capita. These endogenous regressors are instrumented using interactions between the logarithms of village-level chicken and egg prices and predetermined household characteristics, namely poultry capital at the beginning of the production year and household assets per capita. All specifications include the full set of individual, household, and village control variables described in Appendix Table A.3, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. The Kleibergen–Paap LM statistic tests the null hypothesis that the model is underidentified, the Kleibergen–Paap rk Wald F statistic assesses the strength of the excluded instruments under clustered standard errors, and Hansen’s J statistic tests the validity of the overidentifying restrictions. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.8: First-stage estimates for the IV labor-supply models using the broader sample of poultry-producing households observed in at least two survey rounds

	Female		Male	
	$\ln \hat{w}^{*,f}$ (1)	$\ln \hat{y}^*$ (2)	$\ln \hat{w}^{*,m}$ (3)	$\ln \hat{y}^*$ (4)
Log of initial poultry capital $\times \ln P^{chi}$	0.251*** [0.066]	0.040 [0.114]	0.177** [0.088]	-0.009 [0.148]
Log of initial poultry capital $\times \ln P^{egg}$	-0.098 [0.118]	-0.166 [0.206]	0.144 [0.163]	-0.104 [0.256]
Log of per capita asset $\times \ln P^{chi}$	-0.000 [0.070]	0.003 [0.073]	0.088 [0.100]	0.012 [0.081]
Log of per capita asset $\times \ln P^{egg}$	0.121 [0.127]	0.435*** [0.142]	-0.019 [0.178]	0.293* [0.151]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	12644	12644	10287	10287
Partial R^2	0.055	0.021	0.043	0.012
Sanderson–Windmeijer conditional F	73.266	30.932	43.757	30.619

Notes: The table reports the first-stage regressions corresponding to the IV labor-supply models estimated using the broader sample of poultry-producing households observed in at least two BIHS survey rounds. The dependent variables are the logarithm of the household-specific shadow wage and the inverse hyperbolic sine of household-specific shadow income per capita. All specifications include the same control variables and survey-round fixed effects as the corresponding second-stage regressions. Standard errors clustered at the village level are reported in parentheses. The Kleibergen–Paap LM statistic tests and Kleibergen–Paap rk Wald F statistic are shown in Table A.7. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.9: Reduced-form effects of village-level poultry prices on household poultry production

	$\ln Q^{chi}$ (1)	$\ln Q^{egg}$ (2)	$\ln Q$ (3)
$\ln P^{chi}$	0.767** [0.325]		0.623*** [0.218]
$\ln P^{egg}$		0.449 [0.307]	-0.345 [0.347]
Controls	x	x	x
Year FE	x	x	x
Observations	4842	4842	4842
Adj. R^2	0.065	0.029	0.070

Notes: The table reports reduced-form estimates of the relationship between village-level poultry prices and the annual real value of household poultry production for the balanced panel of poultry-producing households observed in all three BIHS survey rounds. The dependent variable is the logarithm of the annual real value of household poultry production, including both chicken and egg output. The key explanatory variables are the logarithms of village-level chicken and egg prices. All specifications include household, village, and market characteristics together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.10: Summary statistics and pairwise correlations of village-level poultry prices

Survey round	Villages	Mean	SD	CV	Min	Max
Real chicken price (usd/kg)						
Round 1	260	2.665	0.353	0.133	1.893	4.309
Round 2	260	2.376	0.266	0.112	1.842	3.562
Round 3	260	1.825	0.298	0.163	1.238	2.776
Real egg price (usd/dz)						
Round 1	260	1.697	0.126	0.074	1.217	2.048
Round 2	260	1.595	0.150	0.094	1.129	1.882
Round 3	260	1.366	0.096	0.070	1.070	1.715
Correlation between chicken and egg prices:						
Round 1	260	0.328				
Round 2	260	0.189				
Round 3	260	0.111				

Notes: The table reports summary statistics for the village-level chicken and egg prices used in the empirical analysis. Village-level prices are constructed as leave-one-out averages of household purchase unit values, with district-round averages used when fewer than two purchase observations are available within a village. The bottom row reports the pairwise correlation between village-level chicken and egg prices for each survey round. The relatively modest correlations indicate that chicken and egg prices exhibit sufficient independent variation to be included simultaneously in the empirical specifications.

Appendix B. Agricultural Household Model

This appendix presents the theoretical framework that motivates the empirical analysis. The model builds on the non-separable agricultural household framework of [Singh et al. \(1986\)](#), [Jacoby \(1993\)](#), and [Skoufias \(1994\)](#). The theoretical framework presented here provides the motivation for the empirical strategy. It does not imply that every theoretical channel is empirically important; rather, the empirical analysis evaluates the extent to which these channels operate in the data.

B.1 Household Optimization

Consider a representative rural household that jointly chooses consumption, production, and labor allocation. Following [Singh et al. \(1986\)](#), household utility depends on consumption and leisure,

$$U = U(C, l_f, l_m), \quad (11)$$

where C denotes household consumption, and l_f and l_m denote leisure of female and male household members, respectively.

Time endowments satisfy

$$T_j = L_j + l_j, \quad j \in \{f, m\}, \quad (12)$$

where L_j denotes labor allocated to productive activities.

B.2 Household Production

Household poultry production is represented by

$$Q = f(L_f, L_m, K; A), \quad (13)$$

where

- Q = value of poultry production (chicken and eggs combined)
- L_f = female family labor,
- L_m = male family labor,
- K = cost of other variable inputs,
- A = value of household fixed productive assets.

The production technology provides the basis for recovering household-specific shadow wages from the marginal value product of family labor.

B.3 Household-Specific Shadow Wages

Under incomplete labor markets, family labor cannot be freely substituted with hired labor. Consequently, the opportunity cost of household labor is determined by its marginal value product rather than the observed market wage.

The household-specific shadow wage for labor type j is

$$w_j^* = \frac{\partial Q}{\partial L_j}, \quad j \in \{f, m\}. \quad (14)$$

Thus, commodity prices influence shadow wages through changes in the marginal value product of family labor.

The household-specific shadow wage depends on both local output prices and the marginal productivity of family labor. Consequently, changes in commodity prices may alter household-specific shadow wages through their effects on the value of marginal products. The direction and magnitude of this relationship depend on the production technology and household adjustment margins and are therefore evaluated empirically.

B.4 Household Shadow Income

Household-specific shadow income equals the value of household production net of production costs,

$$y^* = Q - K + y, \quad (15)$$

where K denotes production costs and y is exogenous additional income.

Household-specific shadow income depends on commodity prices, production levels, production costs, and other household income sources. The extent to which changes in commodity prices translate into changes in household shadow income depends on households' production responses and cost structure and is therefore an empirical question.

For empirical implementation, the production function is estimated using the inverse hyperbolic sine transformation for production inputs that contain zero observations. This transformation affects the estimation procedure but does not alter the underlying theoretical model presented here.

B.5 Household Labor Allocation

Household labor allocation depends on both household-specific shadow wages and household shadow income,

$$L_j = L_j(w_j^*, y^*), \quad (16)$$

which implies

$$\frac{\partial L_j}{\partial P} = \frac{\partial L_j}{\partial w_j^*} \frac{\partial w_j^*}{\partial P} + \frac{\partial L_j}{\partial y^*} \frac{\partial y^*}{\partial P}. \quad (17)$$

The first component represents the substitution effect operating through changes in the opportunity cost of family labor, while the second represents the income effect generated by higher household production income.

B.6 Implications for the Empirical Analysis

Equation 17 provides a theoretical decomposition of the total effect of commodity price shocks on household labor allocation. The decomposition does not impose restrictions on the magnitude or sign of the individual components; rather, it identifies the channels through which commodity price shocks may affect labor allocation under the agricultural household model.

The empirical analysis is designed to evaluate the relative importance of these channels. Specifically, household production functions are estimated to recover household-specific shadow wages and shadow income, after which reduced-form and instrumental-variable specifications are used to assess the extent to which these production-side mechanisms explain observed labor responses to local poultry price shocks.

Appendix C. Inverse Hyperbolic Sine Transformation

C.1 IHS Transformation

Several production inputs used to estimate the household poultry production function contain a substantial number of zero observations. In particular, many backyard poultry-producing households report no male family labor or no expenditures on feed, veterinary medicine, and other purchased inputs during the production year. Estimating the production function using natural logarithms would therefore require excluding these observations or adding arbitrary constants to zero-valued variables, both of which may introduce sample selection or measurement error.

To retain all production observations while preserving the nonlinear relationship between inputs and output, continuous production inputs are transformed using the inverse hyperbolic sine (IHS) transformation. For any non-negative variable x , the transformation is defined as

$$\sinh^{-1}(x) = \ln \left(x + \sqrt{x^2 + 1} \right) \quad (18)$$

The IHS transformation is defined for zero-valued observations and closely approximates the natural logarithm for sufficiently large values. Specifically,

$$\text{IHS}(x) \approx \ln(2x), \quad \text{as } x \rightarrow \infty. \quad (19)$$

Consequently, the transformation retains observations with zero input use while maintaining an interpretation similar to a logarithmic specification over most of the support of the data. The IHS transformation has

therefore become a standard approach for handling zero-valued variables in applied microeconomic research (Bellemare and Wichman, 2020).

In the empirical analysis, the IHS transformation is applied to all continuous production inputs that exhibit zero observations, including female family labor, male family labor, expenditures on feed, veterinary medicine, and other variable poultry-production inputs, the value of the initial poultry stock, and productive household assets. The dependent variable, annual real value of poultry production, is strictly positive for all households included in the estimation sample and therefore enters the production function in natural logarithms.

Using the IHS transformation allows the production function to be estimated using the full balanced panel of poultry-producing households while avoiding the loss of observations associated with logarithmic transformations. The estimated production elasticities are subsequently used to recover household-specific shadow wages and shadow income, as described in Section 5.2. The theoretical agricultural household model presented in Online Appendix B is formulated in levels and is unaffected by this empirical transformation, which is introduced solely for estimation purposes.

C.2 Recovering Shadow Wages under the IHS Specification

As family labor enters the estimated production function using the inverse hyperbolic sine transformation, the corresponding marginal value product must account for the derivative of that transformation. For labor type $j \in f, m$, the estimated production function implies Equation 3

$$\ln \widehat{Q}_{ht} = \dots + \hat{\beta}_j \mathbf{IHS} \left(L_{ht}^j \right) + \dots . \quad (20)$$

Applying the chain rule gives

$$\frac{\partial \widehat{Q}_{ht}}{\partial L_{ht}^j} = \frac{\partial \widehat{Q}_{ht}}{\partial \mathbf{IHS}(L_{ht}^j)} \frac{\partial \mathbf{IHS}(L_{ht}^j)}{\partial L_{ht}^j}. \quad (21)$$

Since,

$$\frac{\partial \mathbf{IHS}(x)}{\partial x} = \frac{1}{\sqrt{x^2 + 1}}, \quad (22)$$

and

$$\frac{\partial \widehat{Q}_{ht}}{\partial \mathbf{IHS}(L_{ht}^j)} = \hat{\beta}_j \widehat{Q}_{ht}, \quad (23)$$

the household-specific shadow wage is

$$\widehat{w}_{j,ht}^* = \hat{\beta}_j \frac{\widehat{Q}_{ht}}{\sqrt{\left(L_{ht}^j\right)^2 + 1}}. \quad (24)$$

This expression is well defined when reported family labor is zero, unlike the standard log–log Cobb–Douglas expression. For sufficiently large positive values of labor input,

$$\sqrt{\left(L_{ht}^j\right)^2 + 1} \approx L_{ht}^j, \quad (25)$$

so Equation (24) approaches the familiar marginal-value-product expression $\hat{\beta}_j \widehat{Q}_{ht} / L_{ht}^j$.

Appendix D. Translog Production Function

The main analysis estimates household-specific shadow wages and shadow income using a Cobb–Douglas production function because it provides a parsimonious representation of household poultry production, facilitates the recovery of marginal value products of family labor, and is well suited to the relatively short household panel. As a robustness check, the production technology is also estimated using a translog specification, which relaxes the Cobb–Douglas assumption of constant output elasticities by allowing for second-order terms and interactions among production inputs.

The translog production function is specified as

$$\ln Q_{ht} = \alpha + \sum_k \beta_k X_{k,ht} + \frac{1}{2} \sum_k \sum_l \gamma_{kl} X_{k,ht} X_{l,ht} + \lambda_t + \mu_h + \varepsilon_{ht}, \quad (26)$$

where Q_{ht} denotes the annual real value of household poultry production, $X_{k,ht}$ includes the same set of transformed production inputs used in the main specification, λ_t denotes survey-round fixed effects, and μ_h represents household fixed effects. As in the baseline model, labor inputs are measured using the inverse hyperbolic sine (IHS) transformation to accommodate zero observations while preserving the panel structure.

Household-specific shadow wages are recovered as the marginal value products of female and male family labor by differentiating the estimated translog production function with respect to the corresponding labor input and evaluating the resulting expression at each household’s observed input levels. Household-specific shadow income is then constructed using the predicted value of poultry production together with household production costs and non-poultry income, following the same procedure described in the main text.

Table A.4 reports the estimated translog production function. The estimated coefficients are broadly consistent with the baseline specification, indicating that the principal findings are not driven by the assumption of a Cobb–Douglas production technology.